Final Project:
8 Bit Multiplier

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ECE 4140-6240: VLSI Design and Simulation
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1. Abstract

This project focuses on the design and implementation of an 8-bit multiplier, a digital circuit capable of multiplying two 8-bit binary numbers to produce a 16-bit result. The design incorporated foundational concepts from digital logic, including the use of AND gates, XOR gates, and full adders, to perform binary arithmetic operations. The project presented several challenges, particularly during the initial design phases, where timing issues in the 8-bit adder caused significant glitches and distorted outputs. A redesign was undertaken, replacing the initial cascaded 4-bit adder structure with a configuration of seven full adders and one half adder, successfully resolving these issues and ensuring stable operation.

Simulations and testing confirmed the correctness of individual components, such as the AND gates and full adders, verified through Design Rule Checks (DRC) and Layout Versus Schematic (LVS). However, final verification of the 8-bit multiplier layout could not be completed due to external constraints, such as licensing issues. Despite this limitation, the project achieved significant milestones in integrating and verifying subcomponents, demonstrating the functionality of the design.

This project builds upon concepts discussed in lectures and labs, highlighting the application of theoretical knowledge to practical digital design challenges. While not introducing groundbreaking innovation, the iterative refinement process provided valuable insights into addressing real-world timing bottlenecks, layout verification challenges, and debugging methodologies. The experience serves as a comprehensive learning opportunity in designing efficient digital systems, bridging academic theory with hands-on implementation.

2. Introduction

Digital multipliers are foundational components in modern computing systems, widely used in applications ranging from arithmetic operations in processors to signal processing in embedded systems. The primary objective of this project is to design and implement an 8-bit multiplier, a digital circuit capable of multiplying two 8-bit binary numbers to produce a 16-bit binary result. This multiplier uses binary arithmetic principles to achieve its functionality, demonstrating how basic logical operations can combine to perform complex computations.

The motivation behind this project stems from the essential role multipliers play in digital systems. Efficient and reliable multipliers are critical for the performance of processors, graphics processing units (GPUs), and digital signal processors (DSPs). Designing such a multiplier provides a deeper understanding of binary arithmetic and circuit optimization techniques, which are integral to digital design. Moreover, the project offers an opportunity to gain hands-on experience with verification and layout processes, essential for ensuring correctness in hardware design.

The scope of the project includes designing the schematic of the 8-bit multiplier, verifying its correctness through simulations, and implementing its physical layout. Key goals include achieving functional accuracy, passing all Design Rule Check (DRC) and Layout Versus Schematic (LVS) checks, and gaining insights into timing and area optimization. By addressing these objectives, the project aims to provide a solid foundation in digital design principles while highlighting the practical challenges of creating high-performance hardware components.

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3. Background and Literature Review

The design of digital circuits, such as an 8-bit multiplier, relies heavily on fundamental building blocks like logic gates and adders. AND, OR, NOR, and XOR gates form the basis of binary arithmetic operations. In a multiplier circuit, AND gates are primarily used to generate the partial products by combining each bit of one operand with every bit of the other. XOR gates are essential in constructing adders, as they compute the sum in binary addition by identifying when inputs differ. OR and NOR gates are less commonly employed in the core computation of multipliers but play crucial roles in control logic and error-checking mechanisms within digital systems.

Half and full adders are integral components in the summation of partial products in a multiplier. A half adder, which combines an XOR gate for sum generation and an AND gate for carry output, is used when there are no incoming carries. Full adders extend this functionality by incorporating an additional input to account for carry bits from previous stages, enabling their use in cascaded configurations for multi-bit addition. Together, these components enable the effective implementation of binary arithmetic required for multiplication.

Existing solutions to digital multiplication span from basic shift-and-add algorithms to optimized array and Wallace tree multipliers, which aim to improve speed and area efficiency. Array multipliers use a grid of adders and logic gates to sum partial products in a structured manner, while Wallace tree multipliers employ hierarchical grouping of adders to reduce the depth of addition stages. While effective, these designs often prioritize speed or area optimization, leaving room for improved trade-offs between performance and resource usage.

This project addresses several gaps in existing designs. Traditional multipliers often focus exclusively on performance metrics, sometimes at the expense of simplicity or design verification ease. By emphasizing both functional accuracy and comprehensive verification through Design Rule Check (DRC) and Layout Versus Schematic (LVS) processes, this project aims to deliver a design that balances performance with design robustness. Additionally, the project incorporates learning objectives that highlight the practical challenges of physical implementation, bridging the gap between theoretical circuit design and real-world constraints.

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4. Innovation

The 8-bit multiplier design was not an entirely new innovation but rather an application of well-established digital design principles and methodologies. This project built upon the foundational concepts covered in lectures and labs, such as binary arithmetic, the use of logic gates, and the implementation of adders. By leveraging these concepts, we aimed to create a robust and efficient 8-bit multiplier capable of reliably performing multiplication operations. The primary focus was to synthesize and integrate these components into a cohesive design while addressing practical challenges such as timing delays, layout verification, and simulation accuracy.

Compared to existing solutions, our design followed a conventional approach but incorporated refinements based on the specific challenges encountered during development. For instance, the initial use of cascaded 4-bit adders for the 8-bit adder was an exploration of a modular design strategy. However, this approach introduced significant timing issues, leading us to adopt a more integrated design using seven full adders and one half adder. This adjustment not only resolved the timing glitches but also improved the overall reliability of the circuit. While this solution did not deviate significantly from standard practices, it demonstrated a commitment to iterative improvement and practical problem-solving.

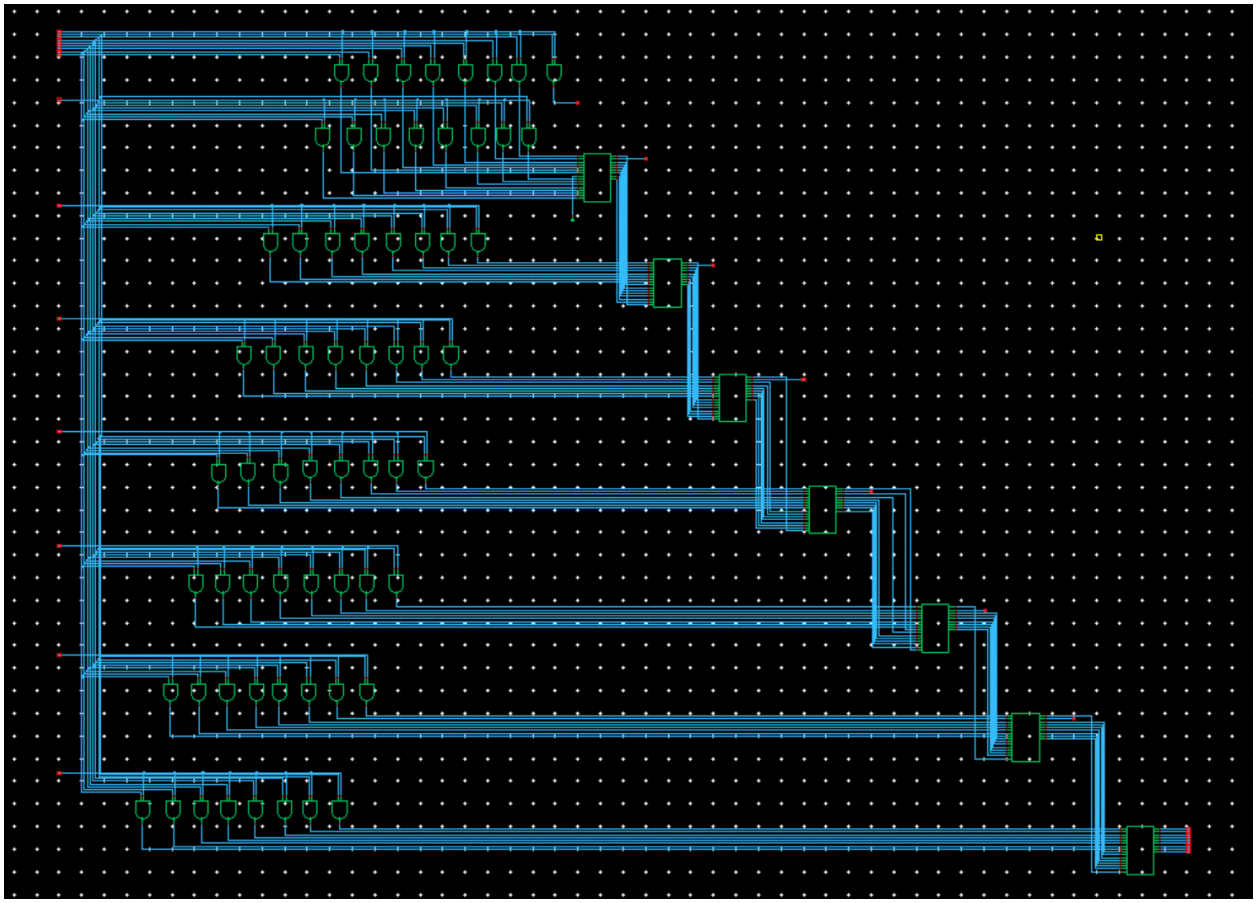
The project's unique contribution lies in its detailed analysis and resolution of real-world challenges encountered during the design and implementation process. While the multiplier itself is a common digital circuit, the lessons learned from debugging timing issues, verifying layouts, and optimizing component integration provide valuable insights for future designs. This iterative refinement process, grounded in theoretical knowledge and practical application, underscores the importance of bridging classroom concepts with hands-on problem-solving in digital design projects.

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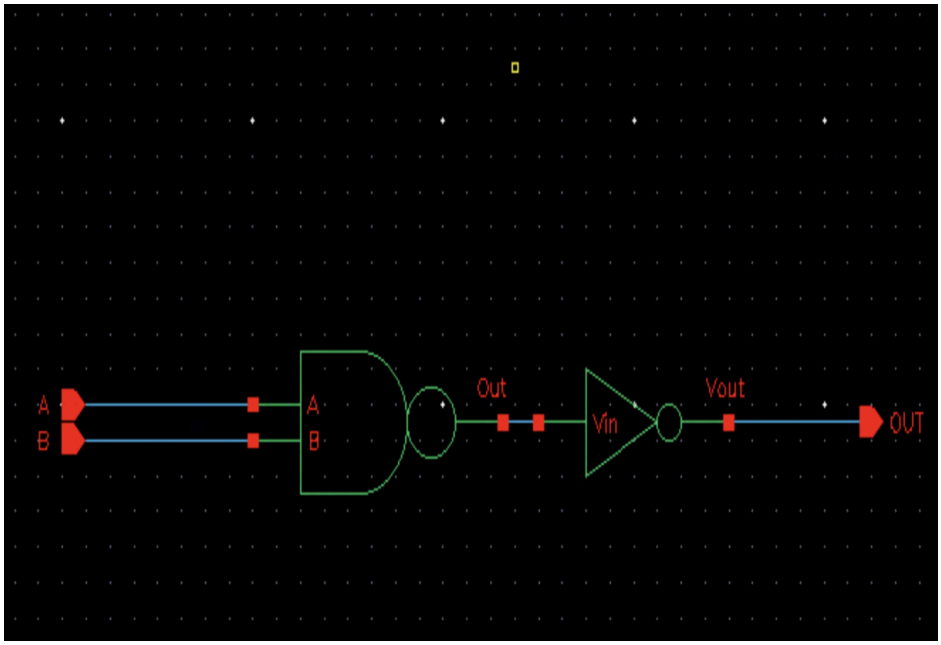
5. Design and Implementation

a) **Schematic Design**

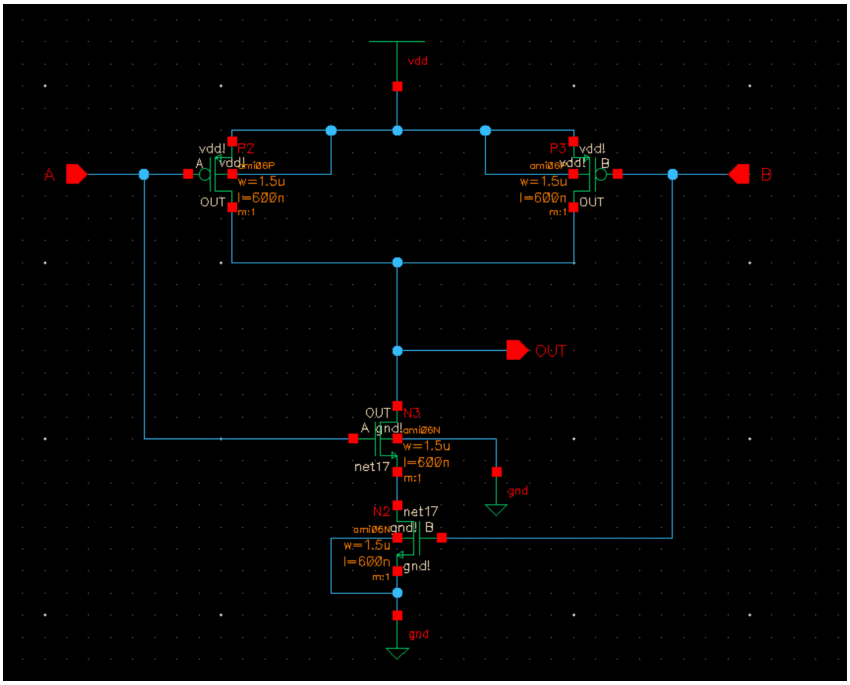
Full schematic of 8-Bit Multiplier using AND Gates as well as 8-Bit Adders



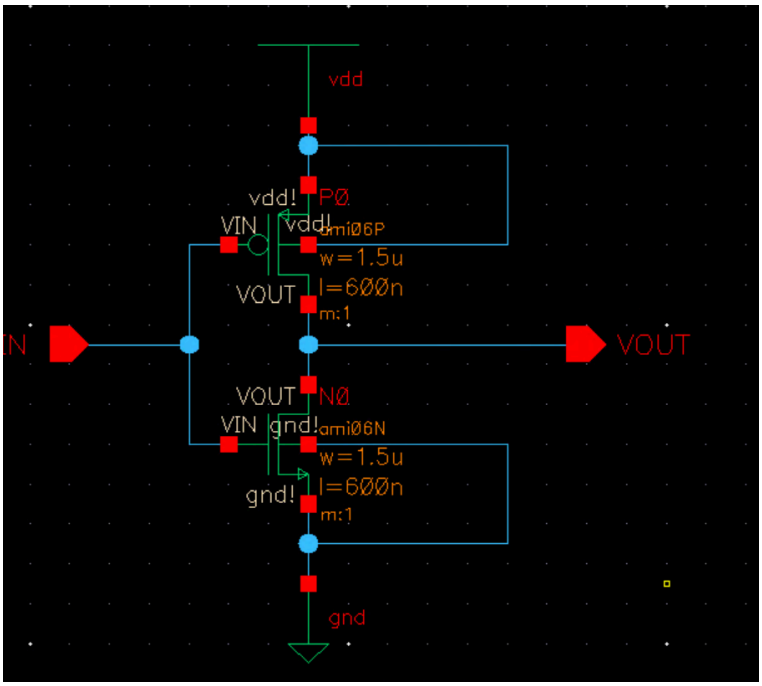
AND Gate



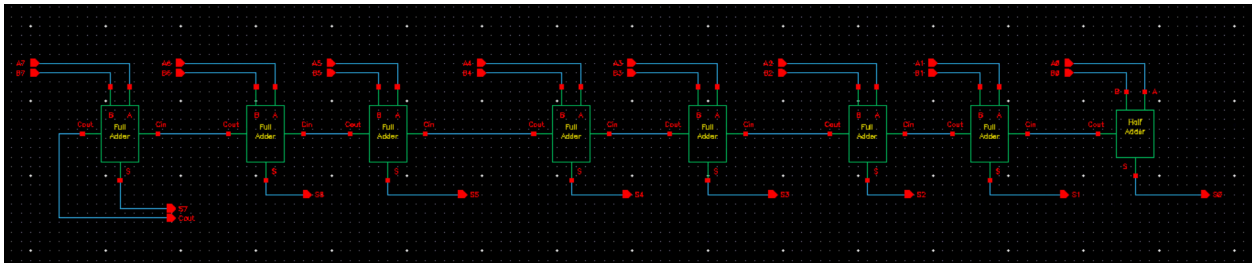
NAND Gate



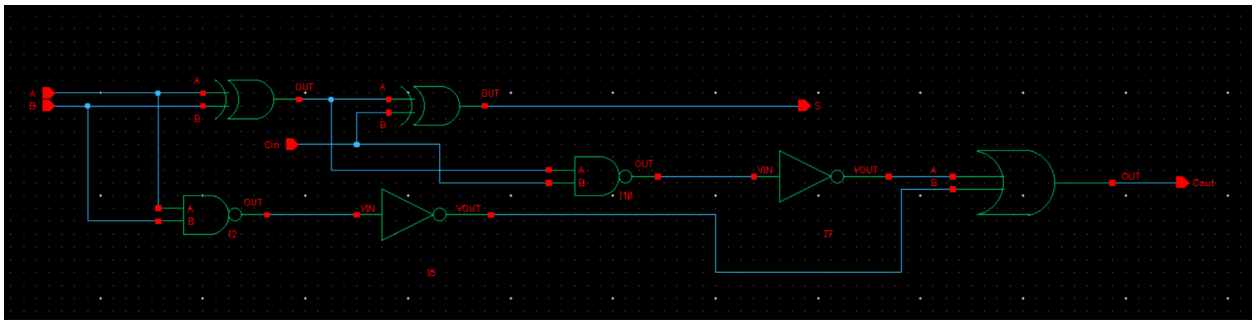
Inverter Gate



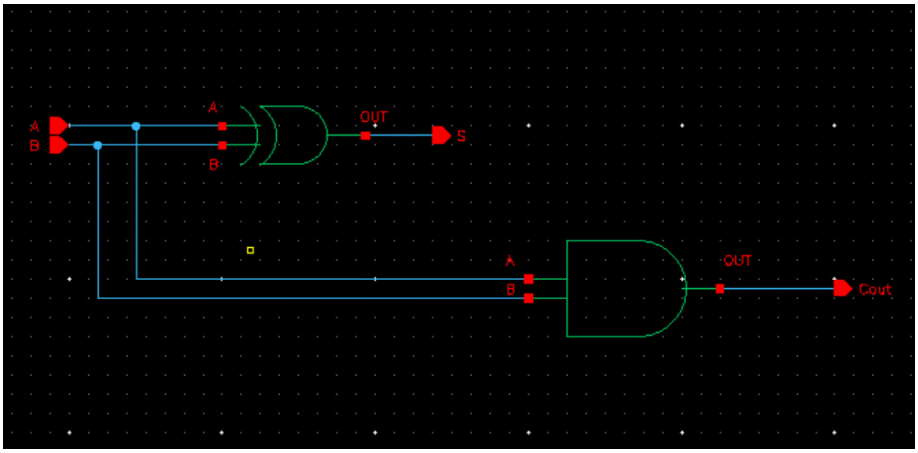
8-Bit Adder



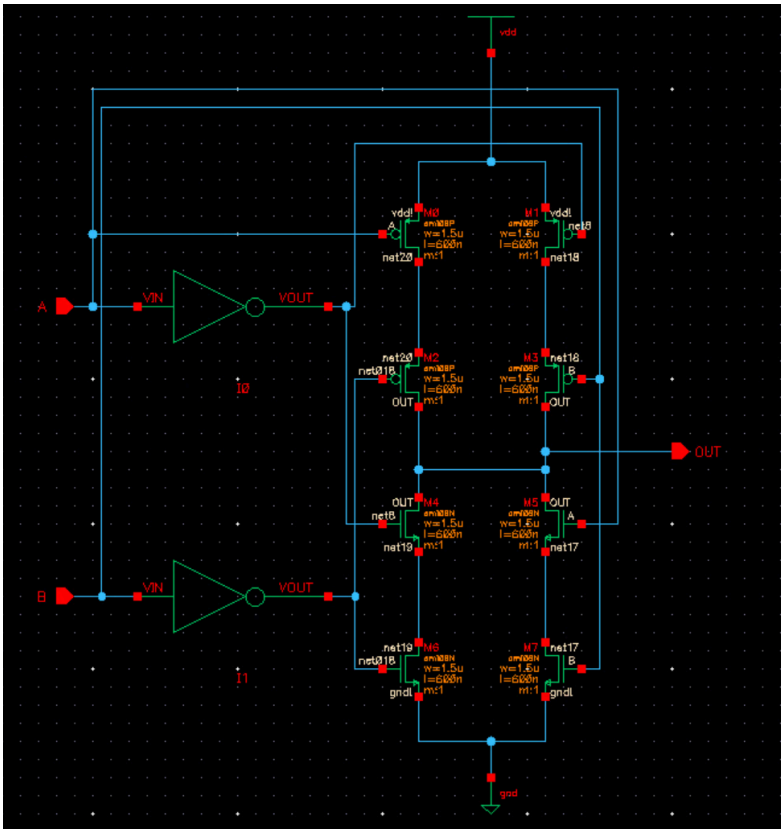
Full Adder



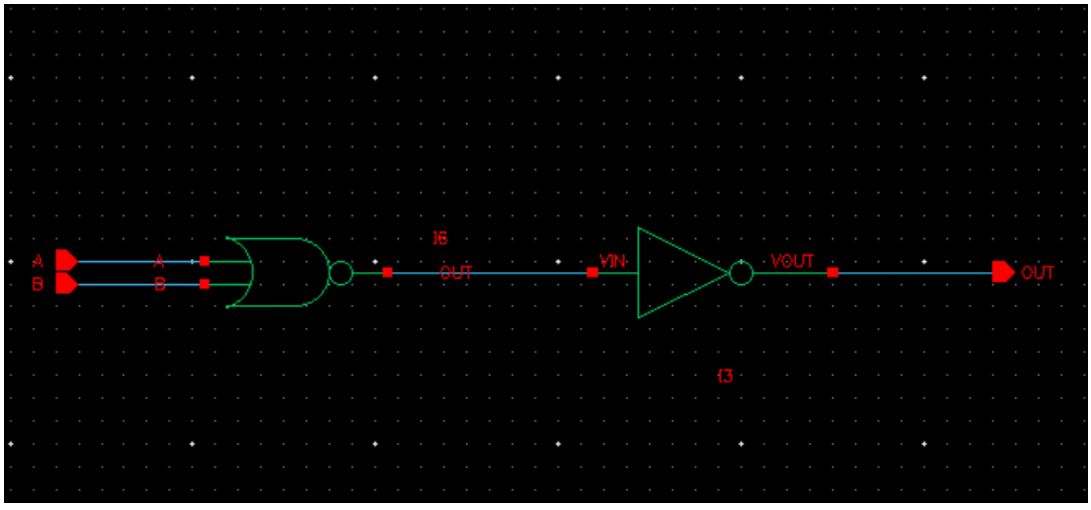
Half Adder



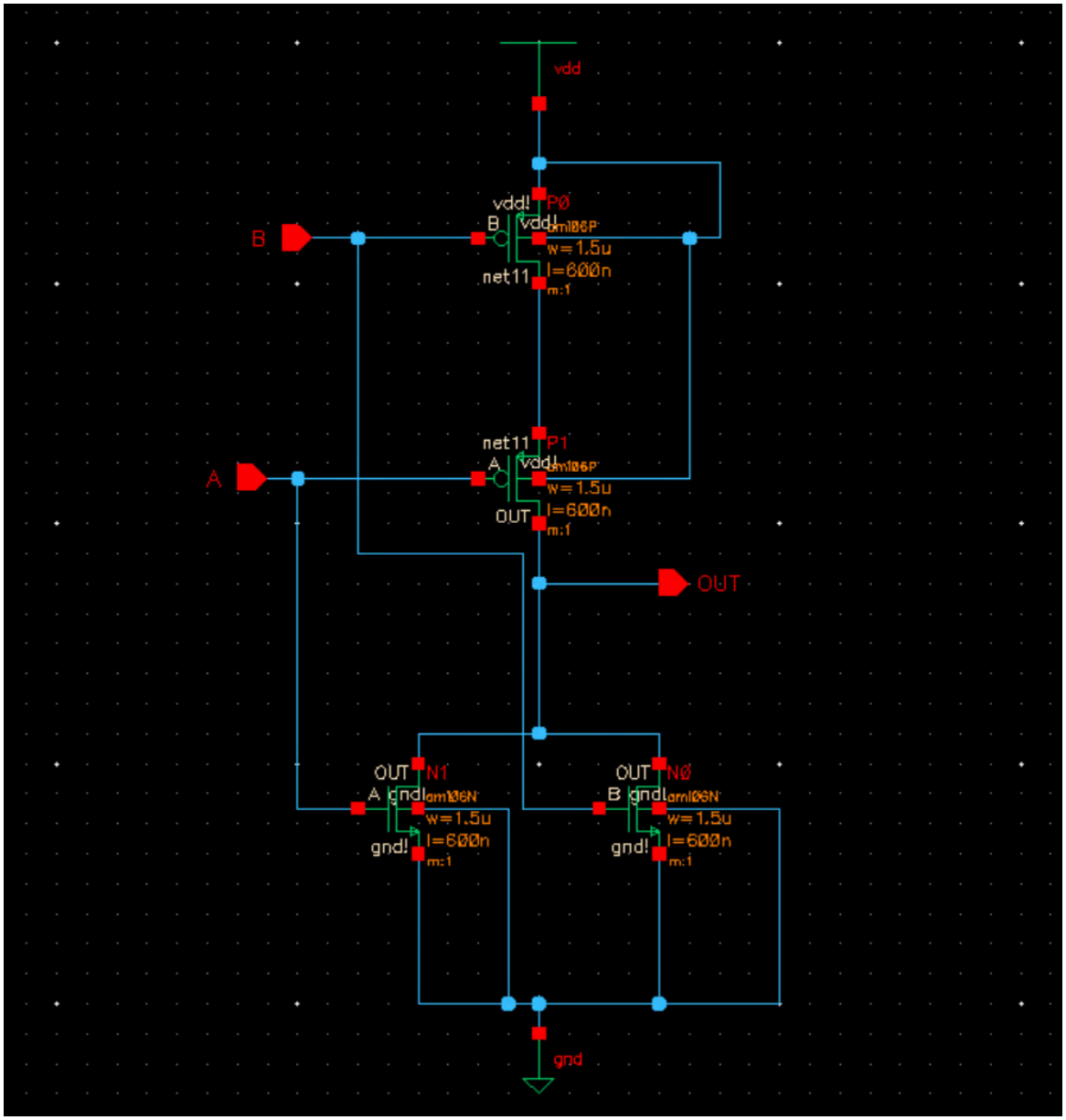
XOR Gate



OR Gate

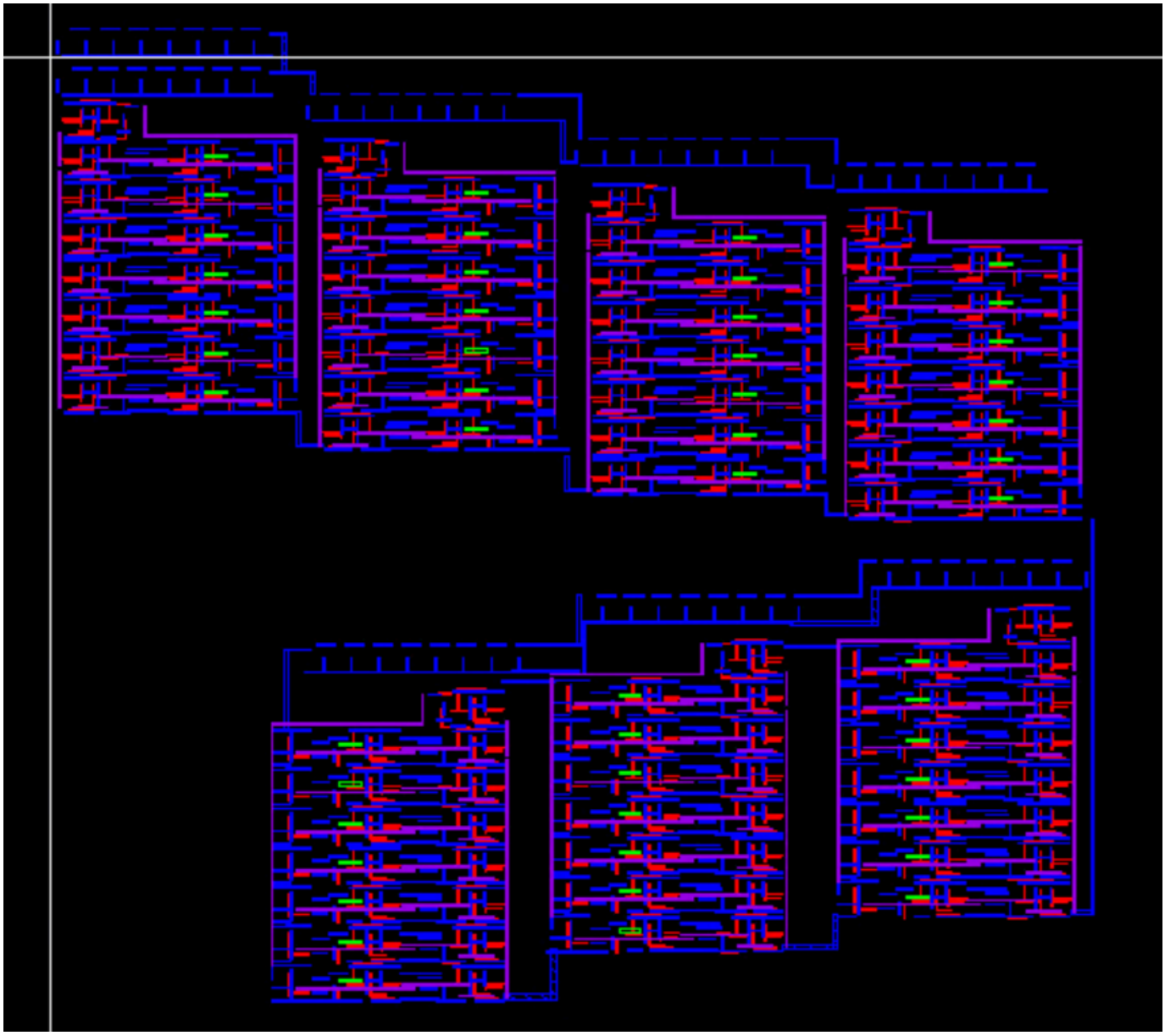


NOR Gate



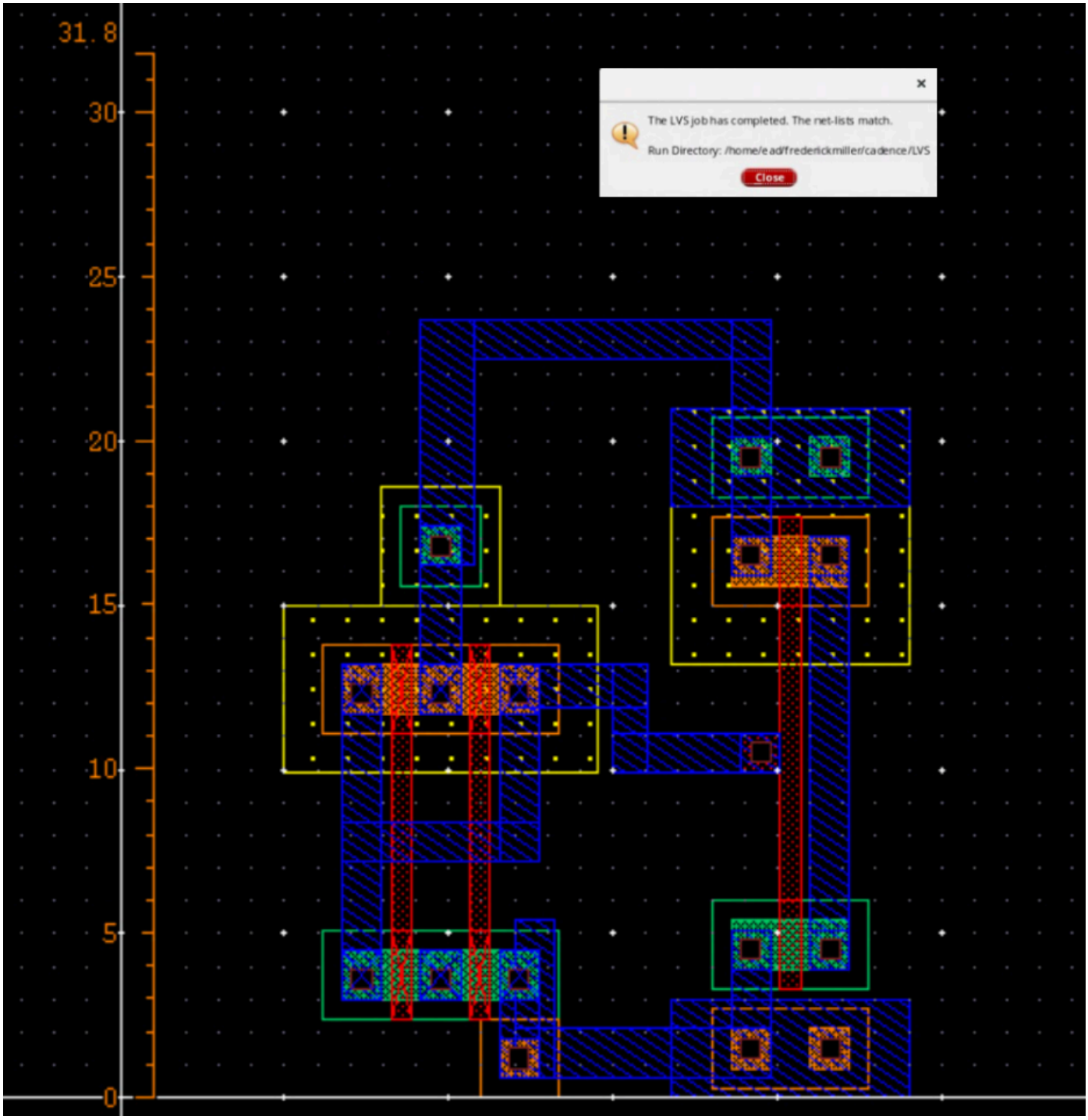
b) Verification and Layout Design

8-Bit Multiplier

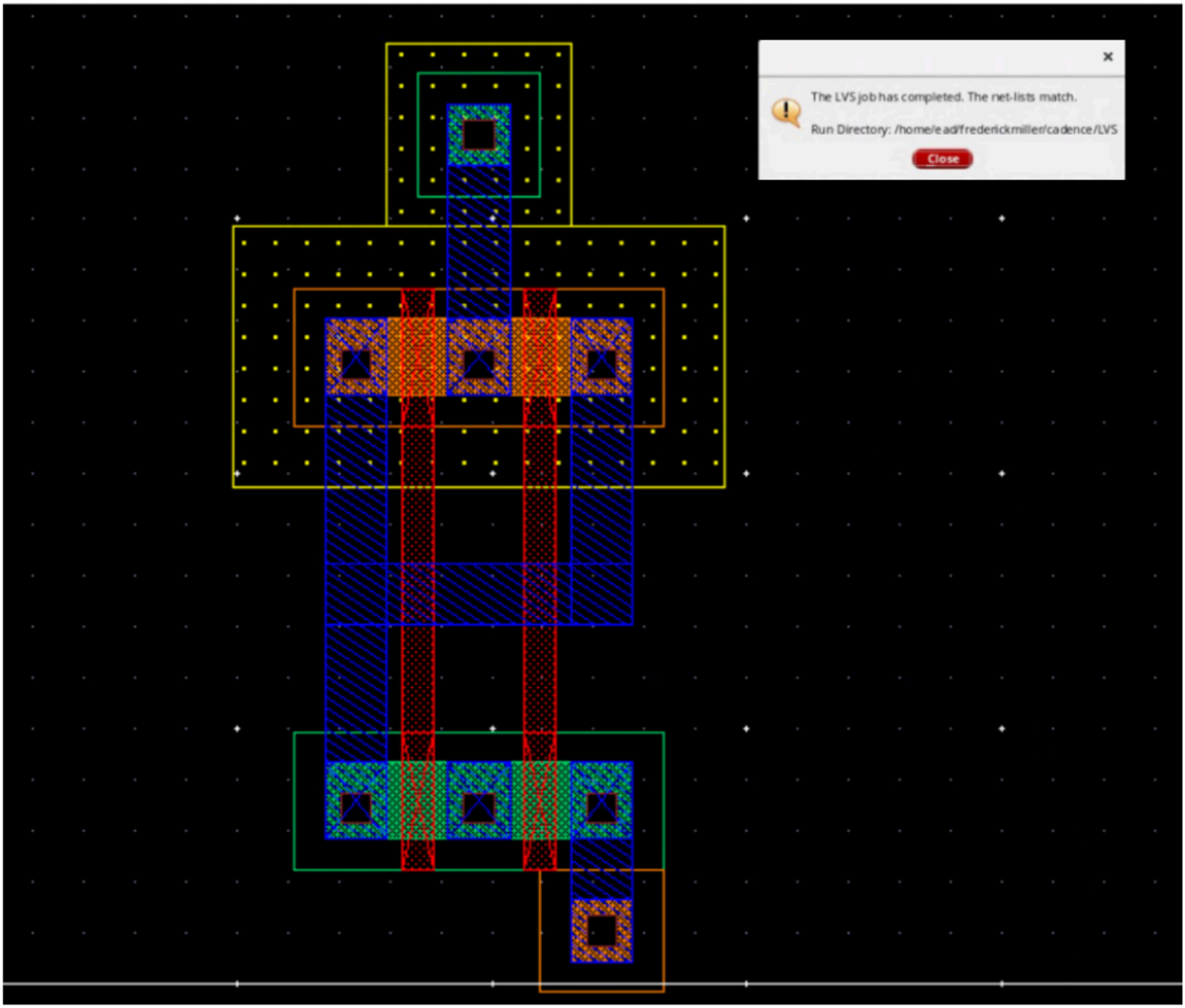


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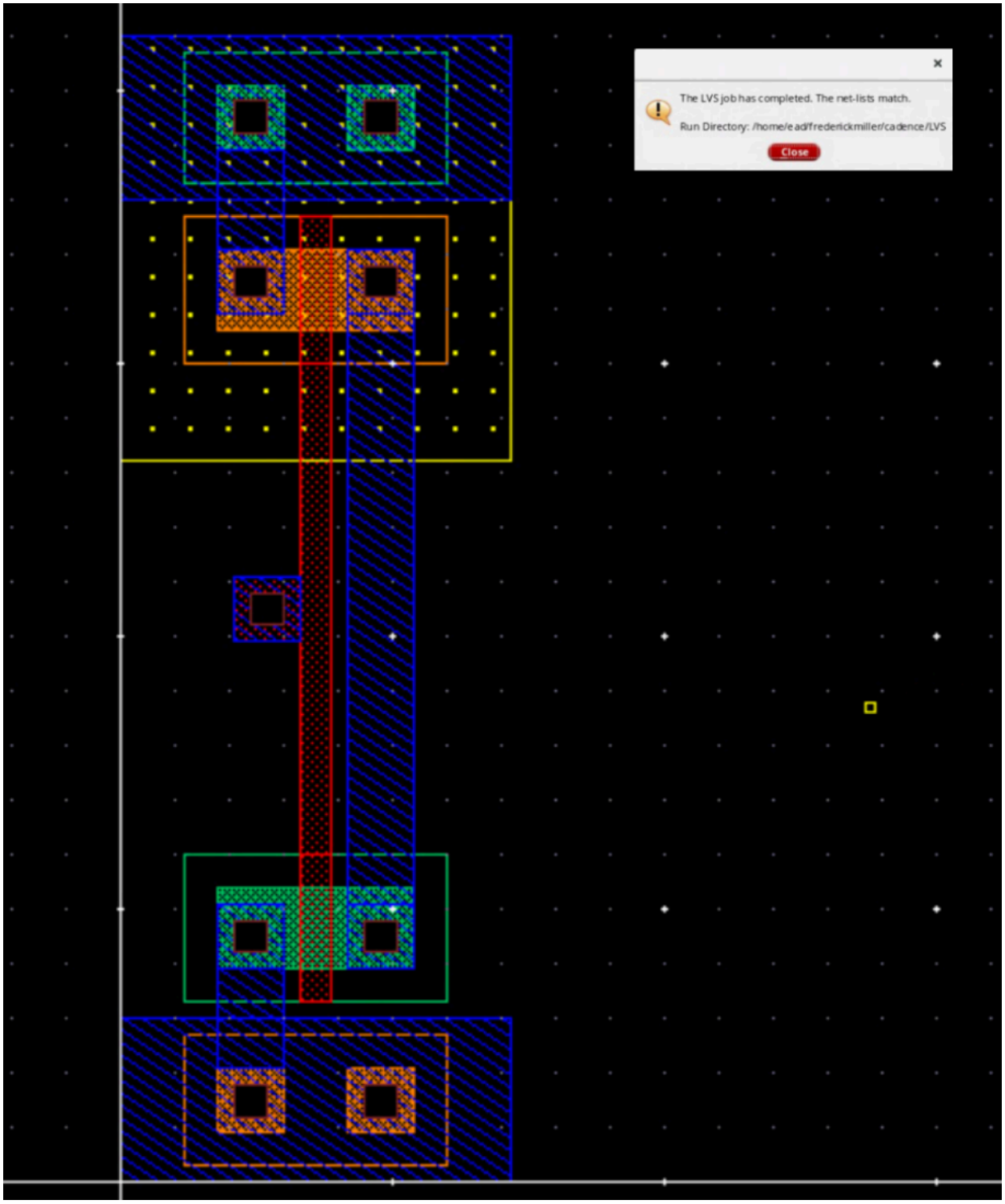
AND Gate


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The LVS job has completed. The net-lists match.
Run Directory: /home/ead/frederickmiller/cadence/LVS
Close

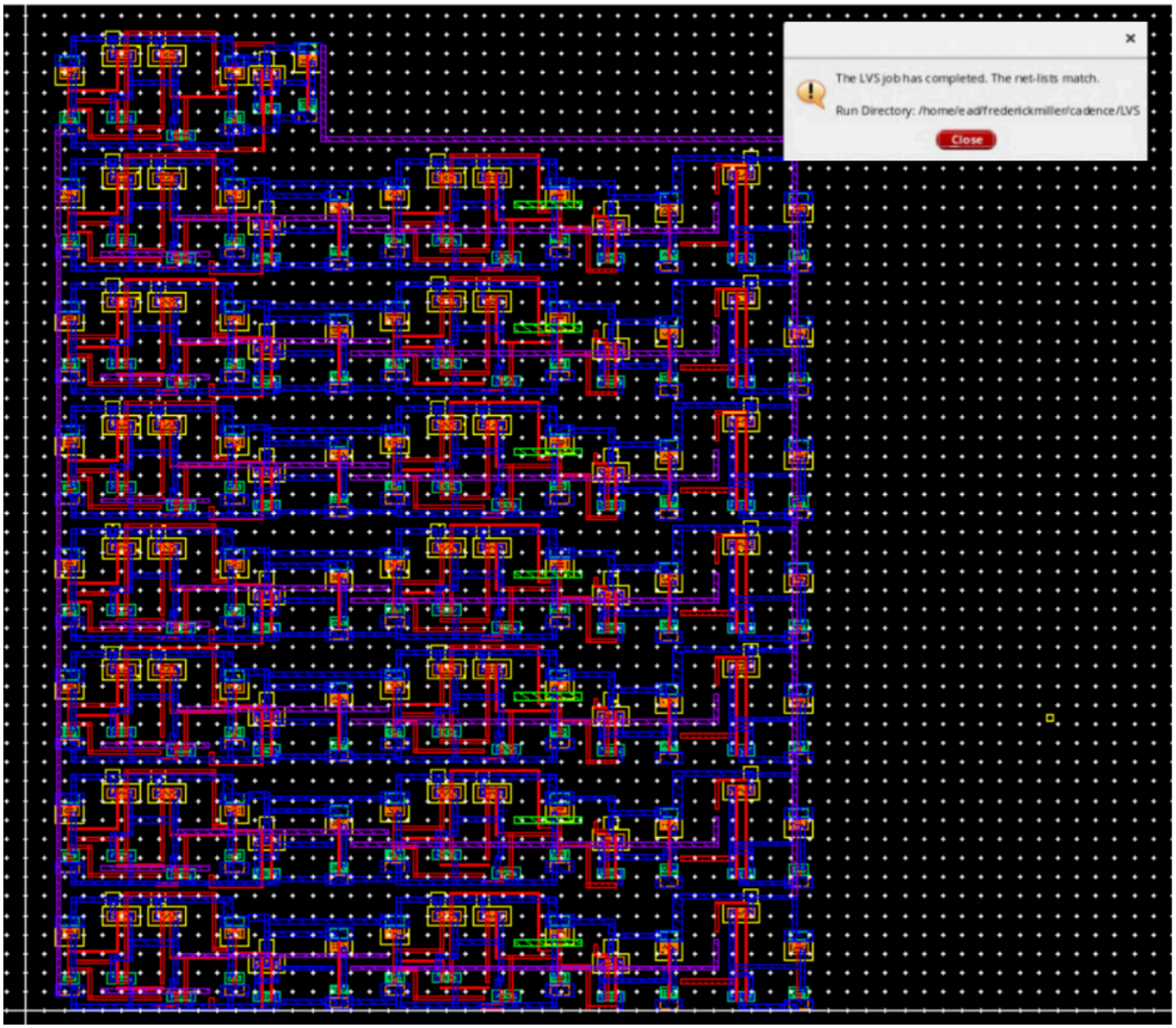
NAND Gate


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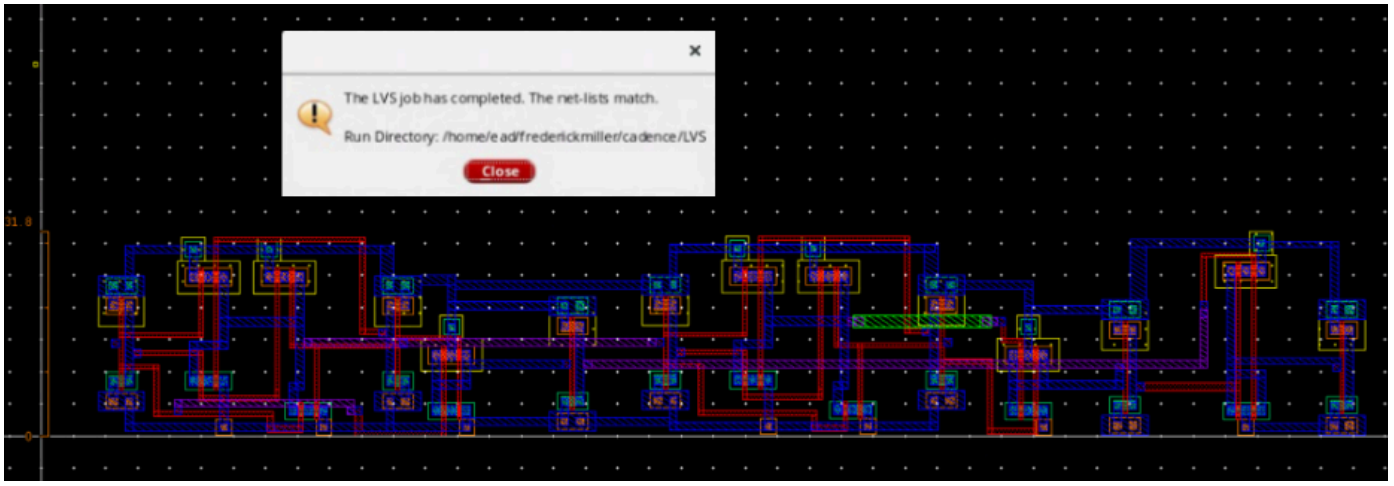
Inverter Gate



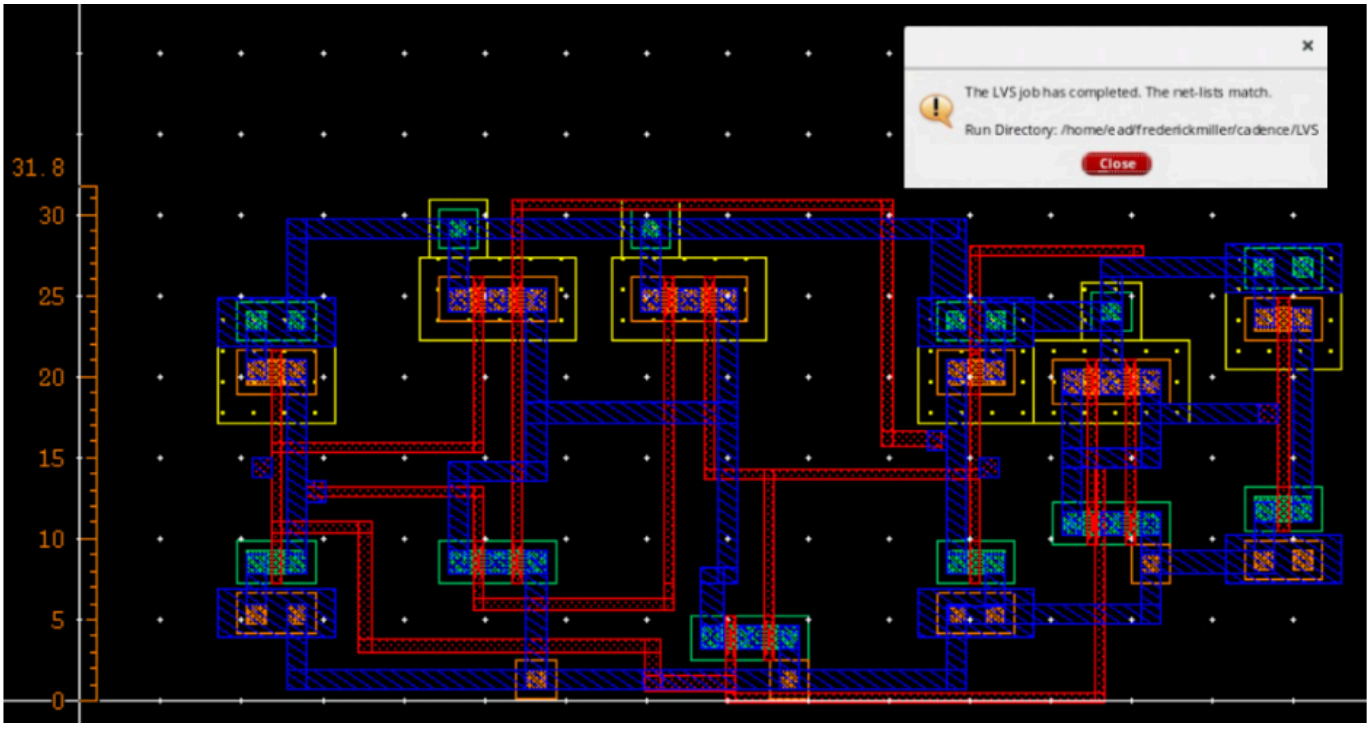
8-Bit Adder


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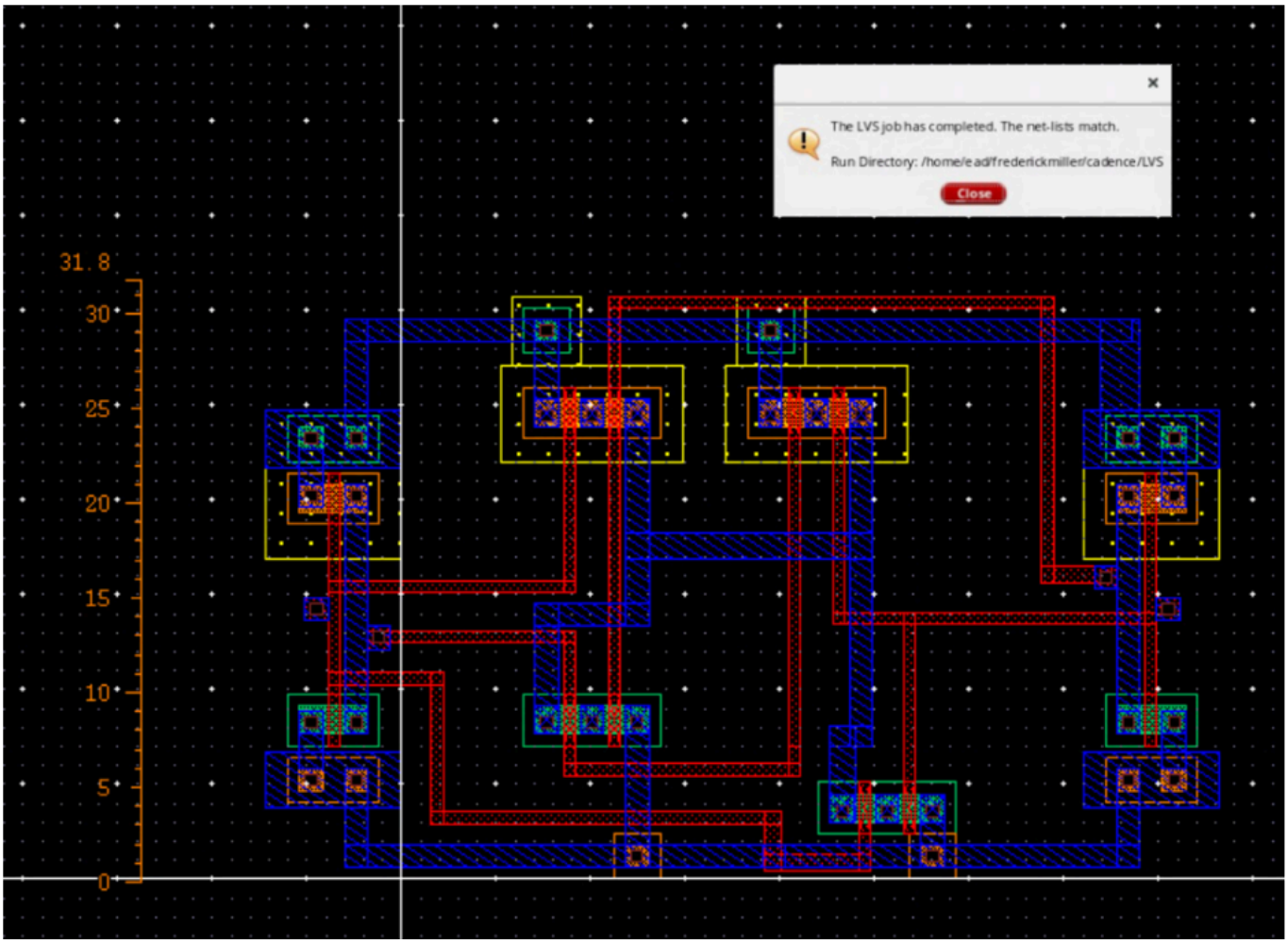
Full Adder



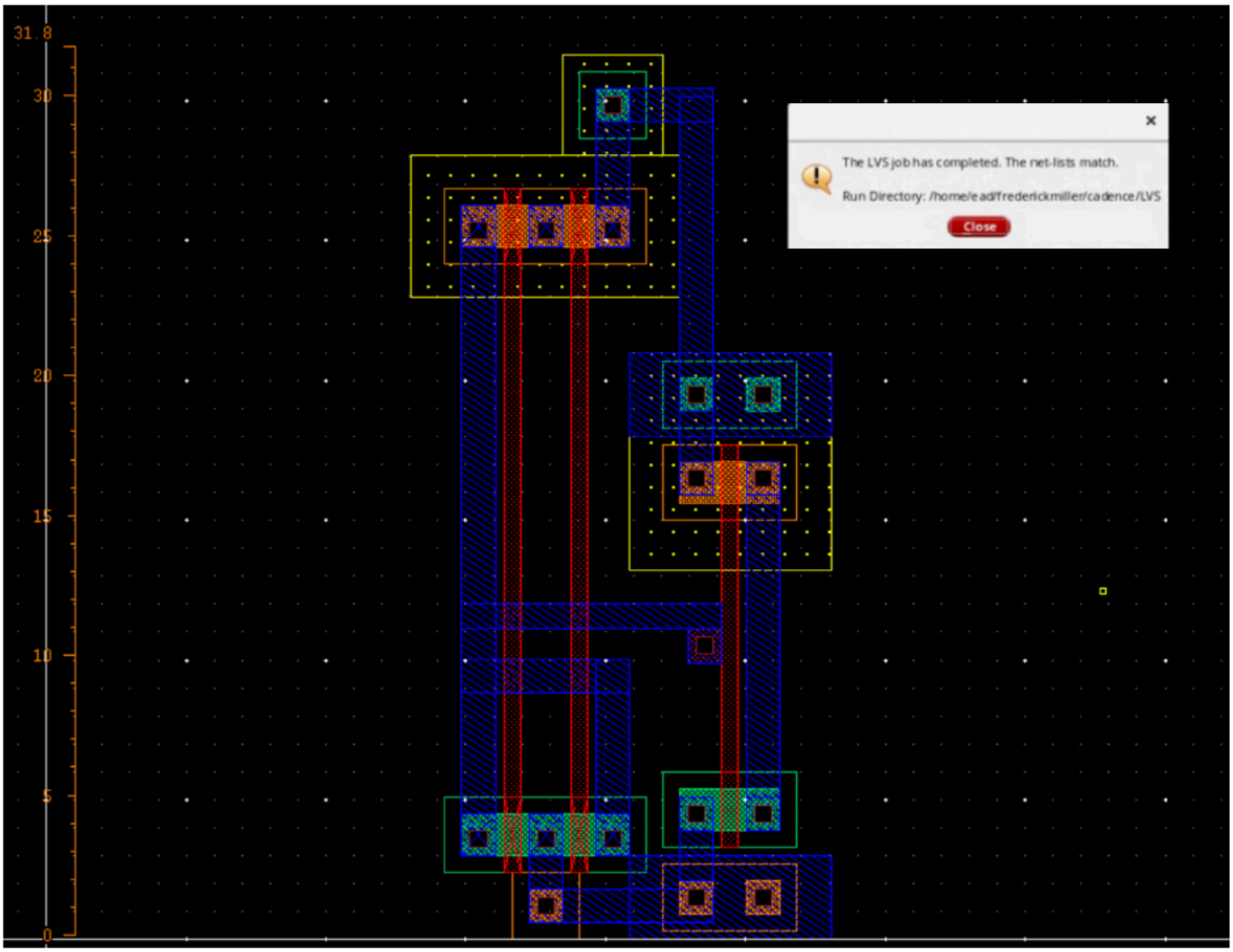
Half Adder



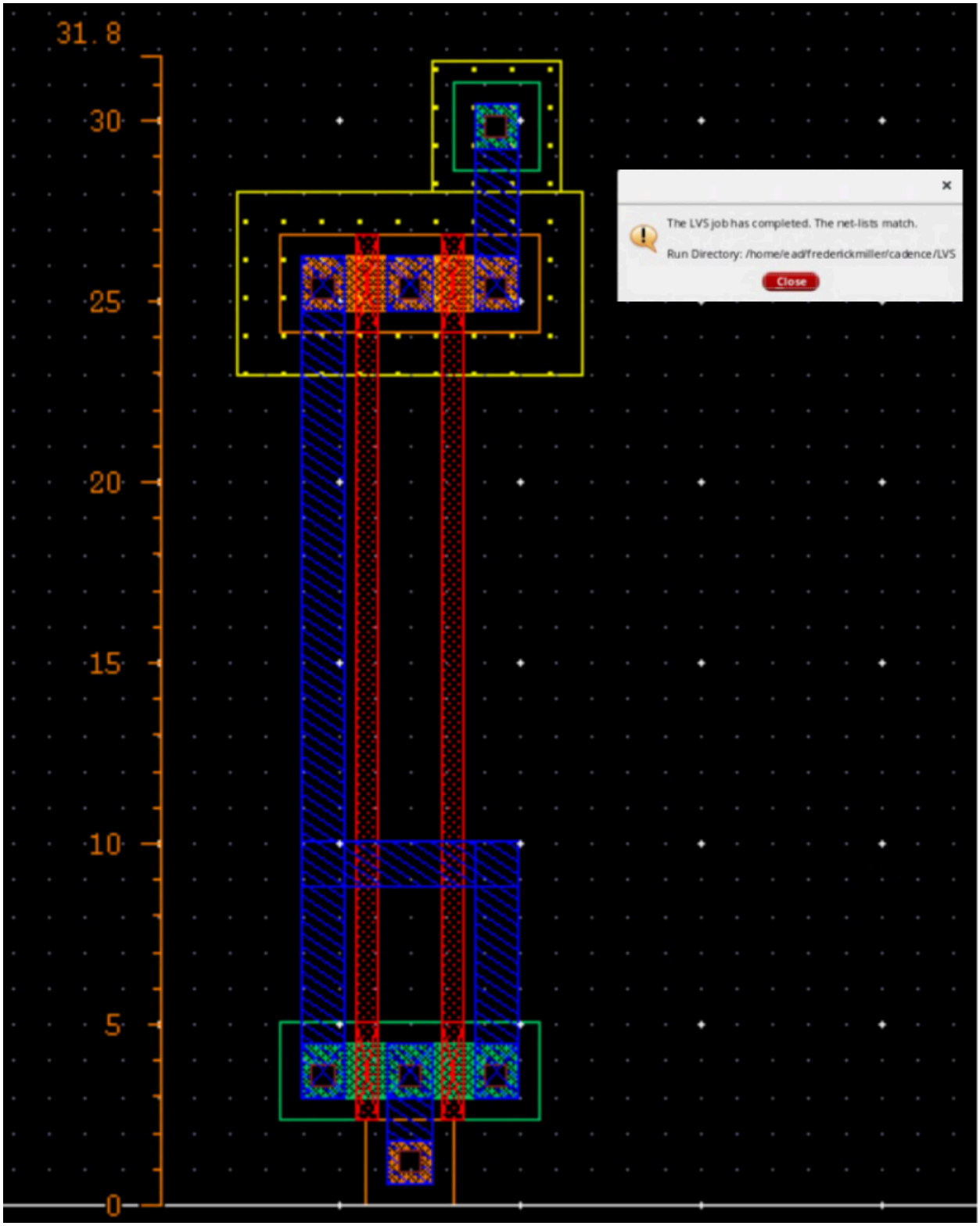
XOR Gate


The LVS job has completed. The net-lists match.
Run Directory: /home/ead/frederickmiller/cadence/LVS
Close
31.8
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OR Gate


31.8
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10
5
0
The LVS job has completed. The net-lists match.
Run Directory: /home/ead/frederickmiller/cadence/LVS
Close

NOR Gate

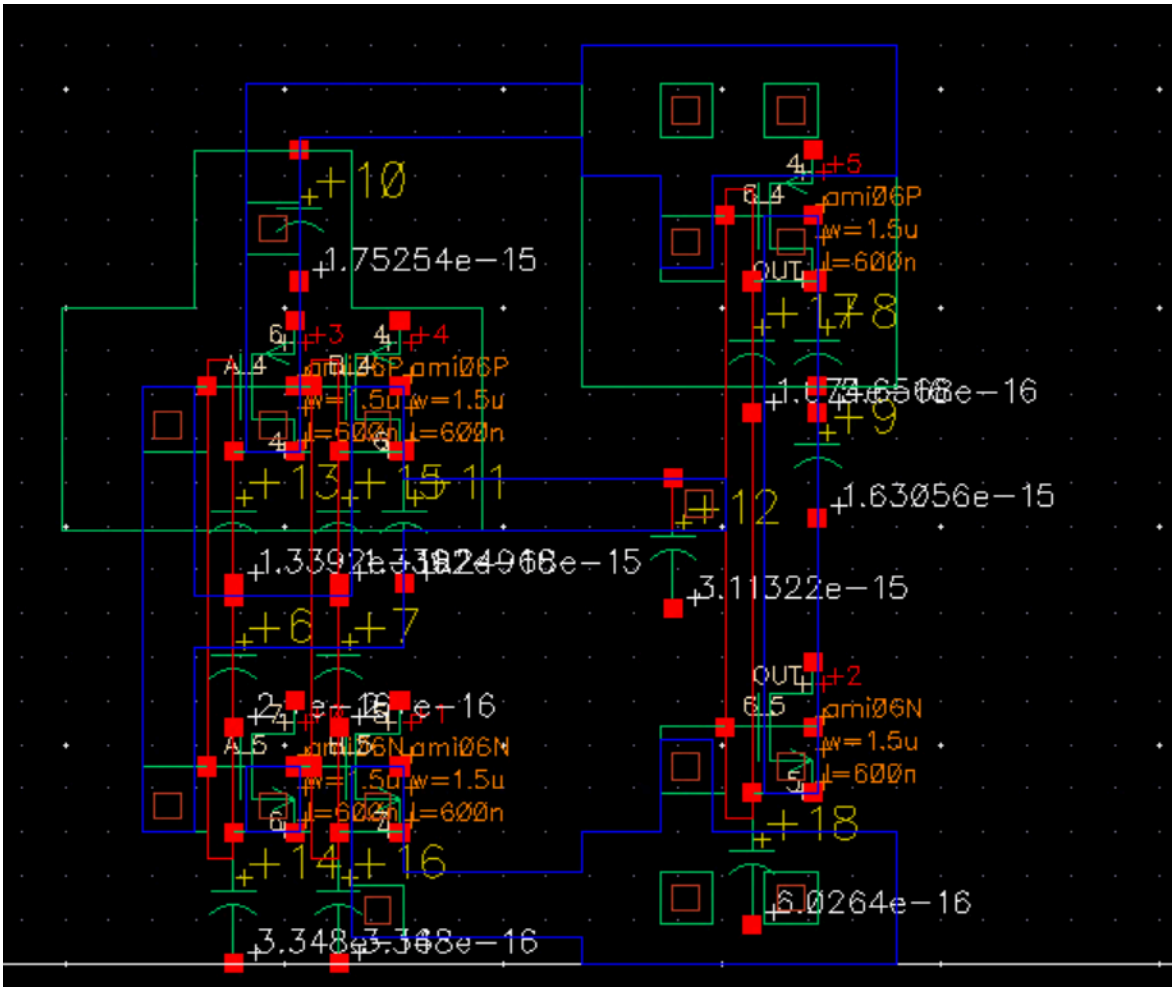

31.8
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Close

c) Extracted Parasitic View

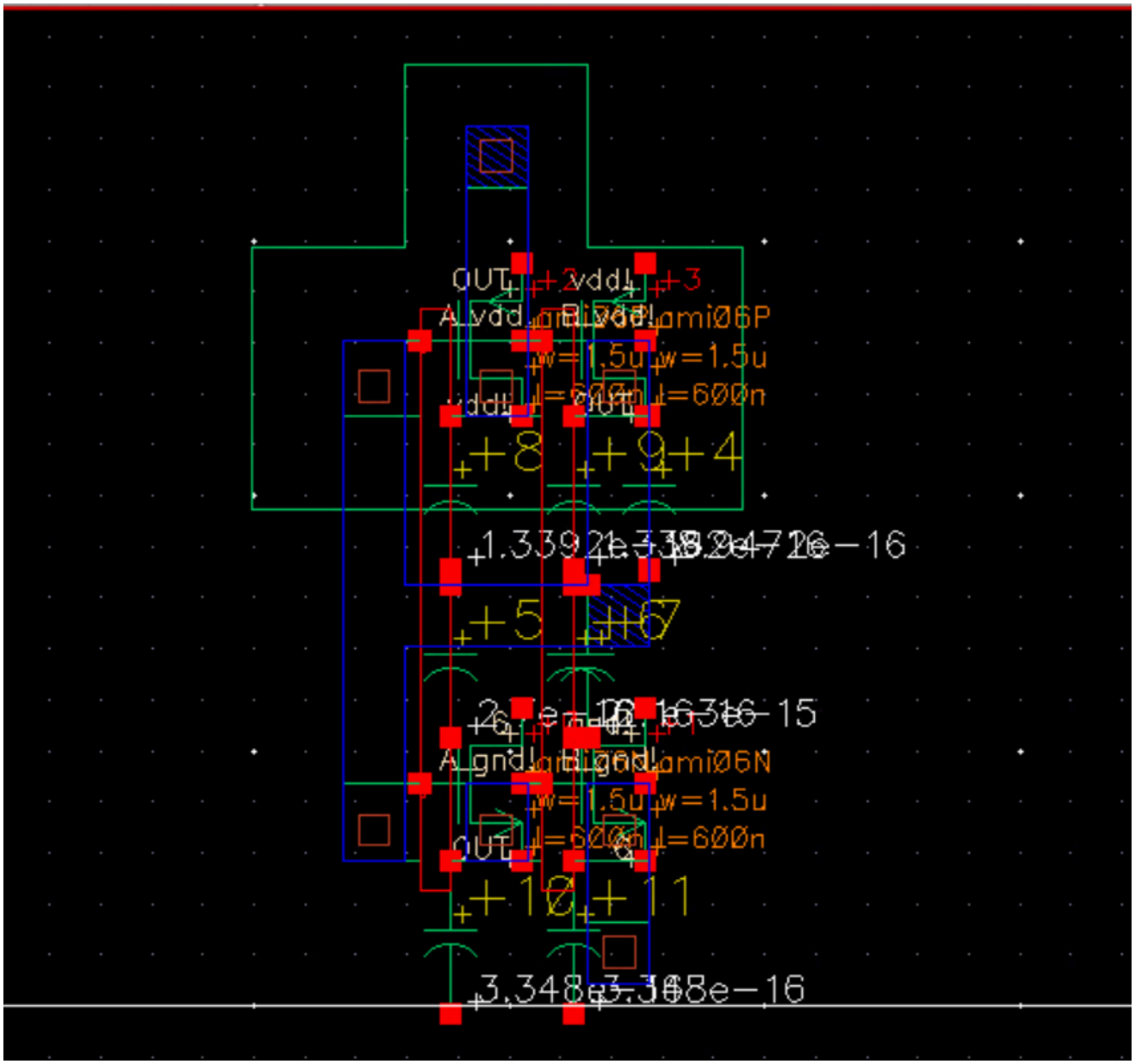
8-Bit Multiplier

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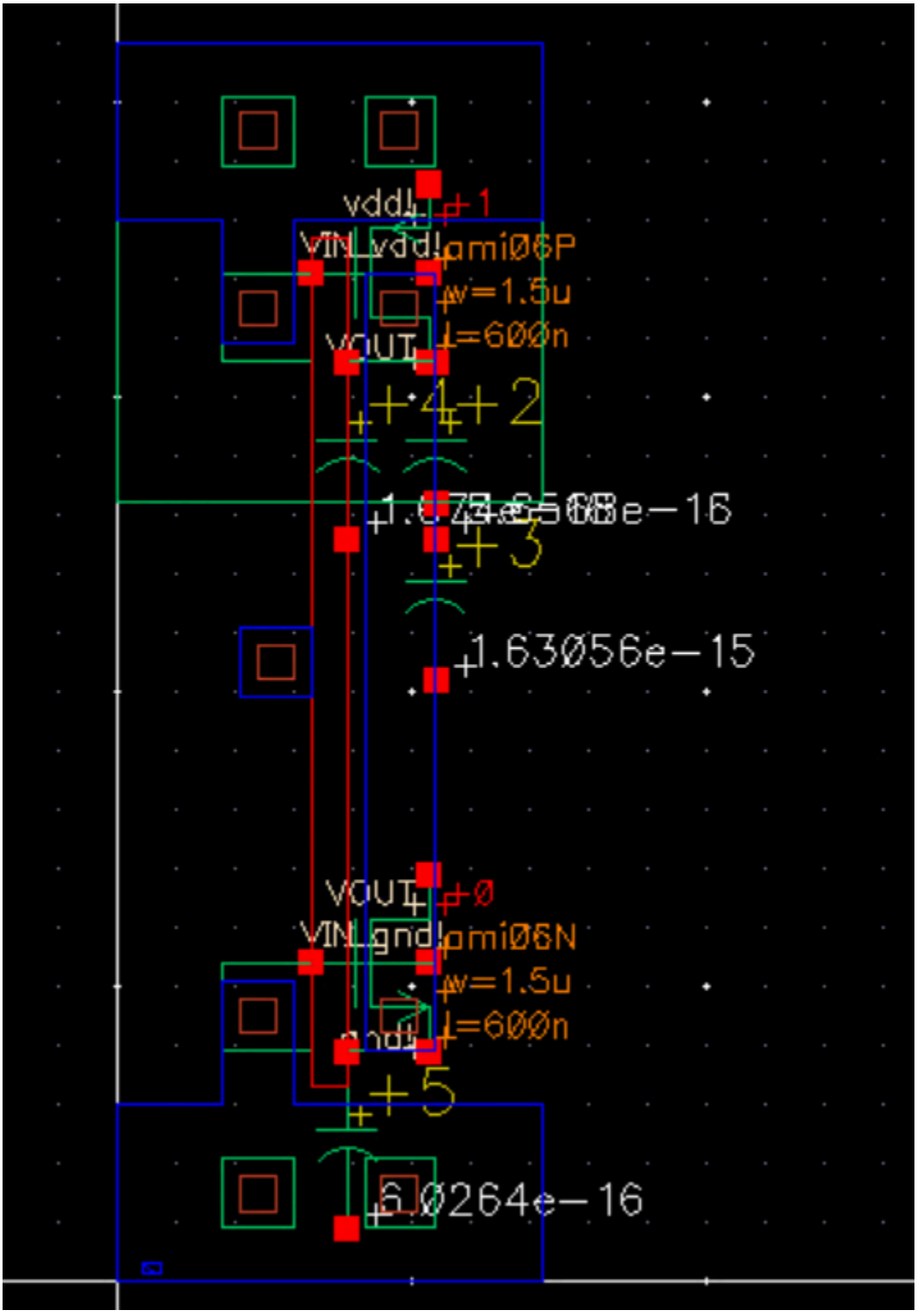
AND Gate



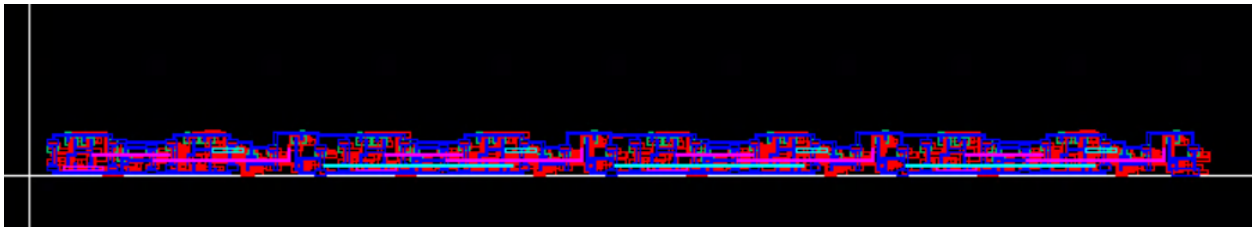
NAND Gate


OUT
vdd!
A
vdd
gnd!
ami06P
w=1.5u
w=1.5u
l=600n
l=600n
+8
+9
+4
+5
+7
A
gnd!
gnd!
ami06N
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w=1.5u
l=600n
l=600n
OUT
+10
+11

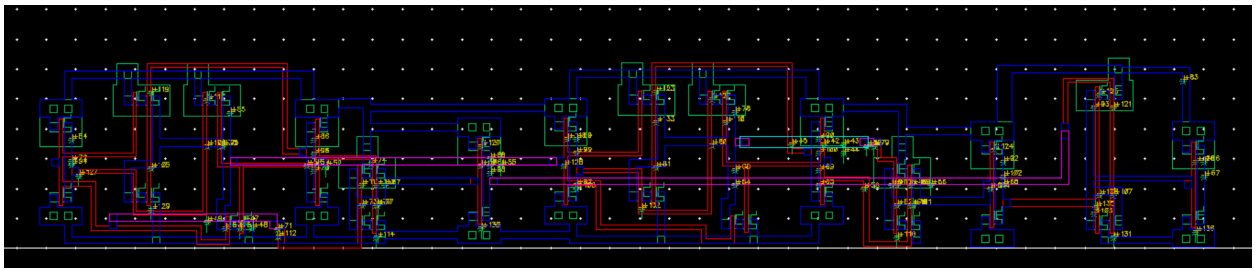
Inverter Gate



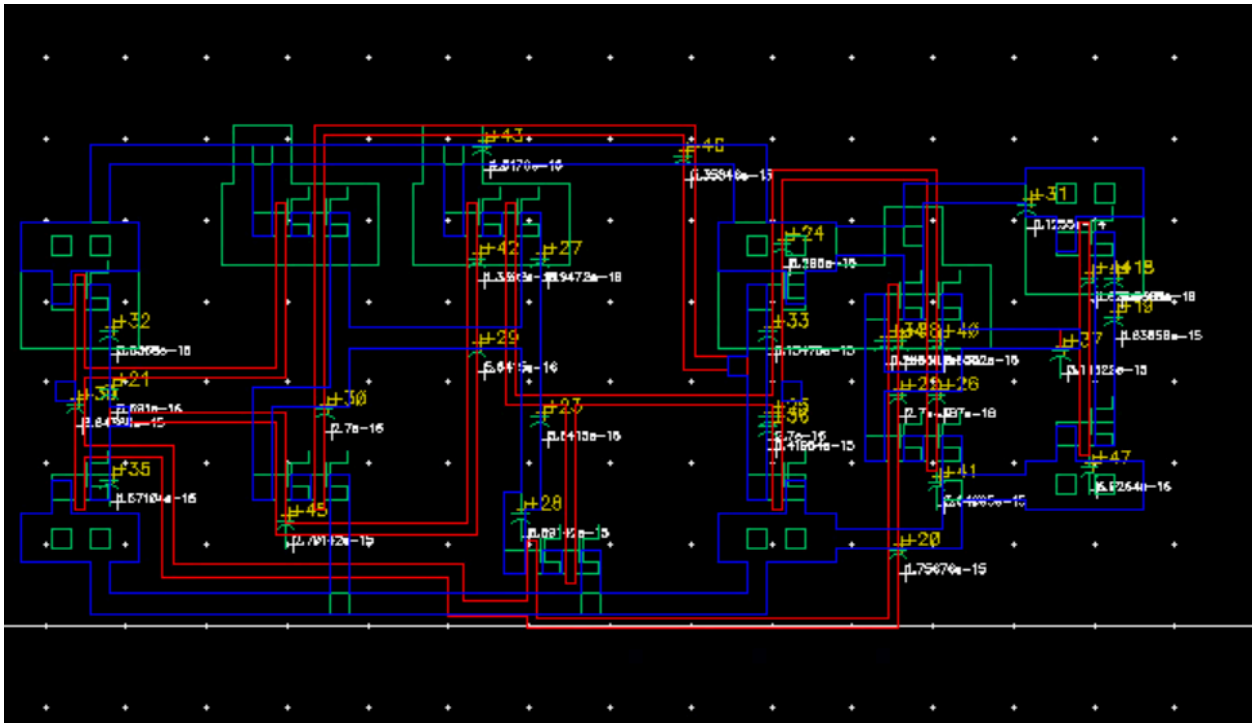
8-Bit Adder



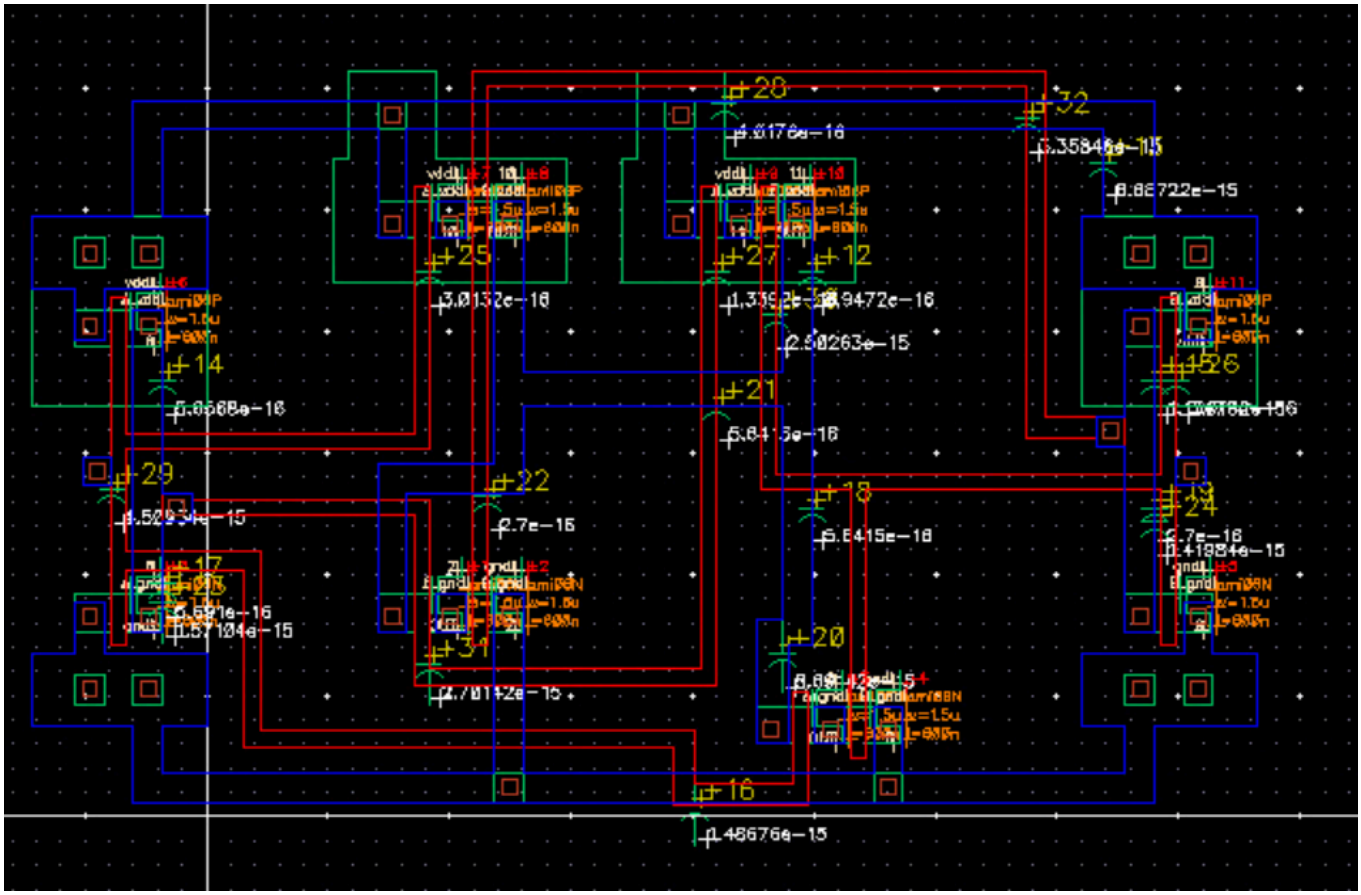
Full Adder



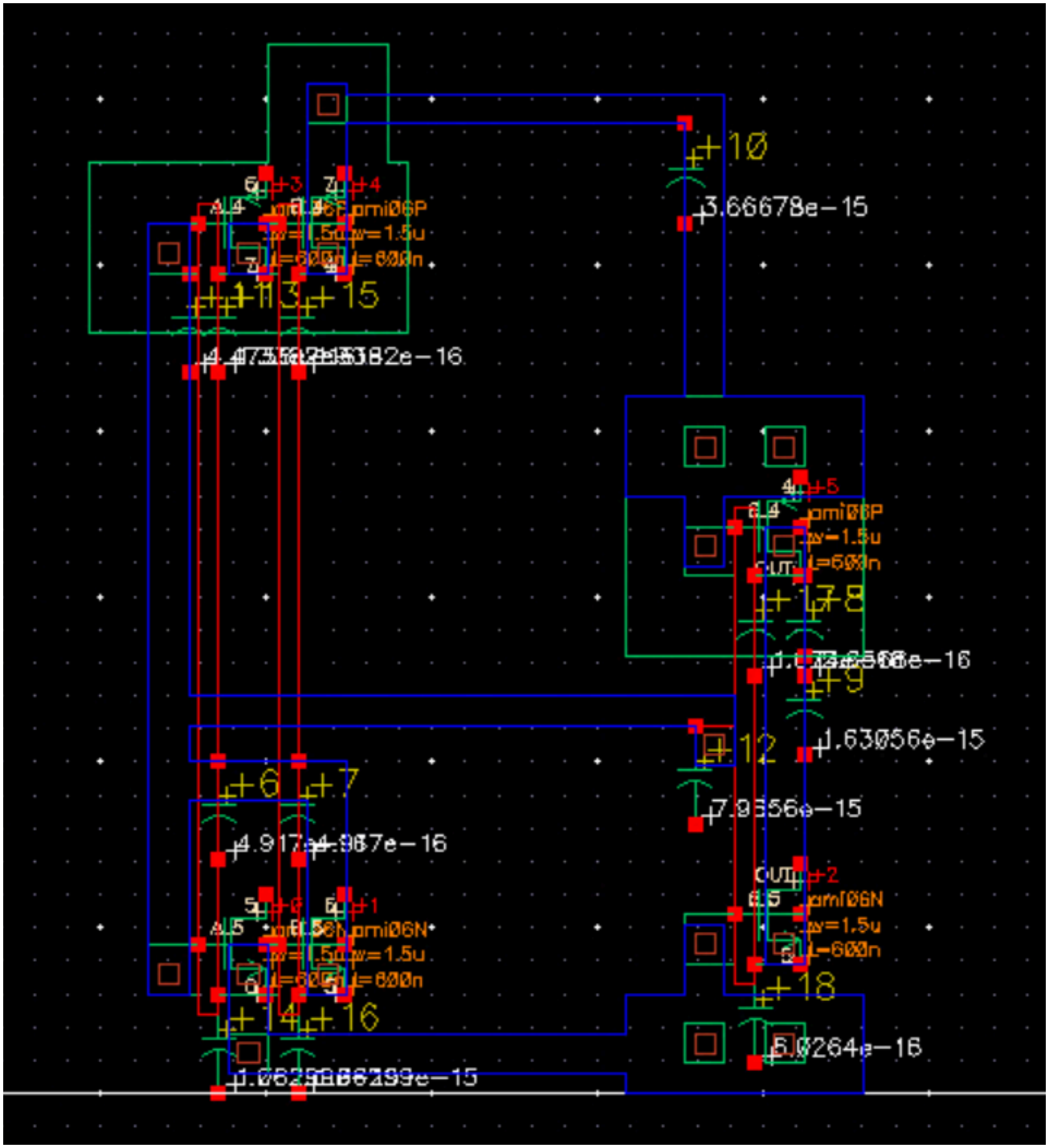
Half Adder



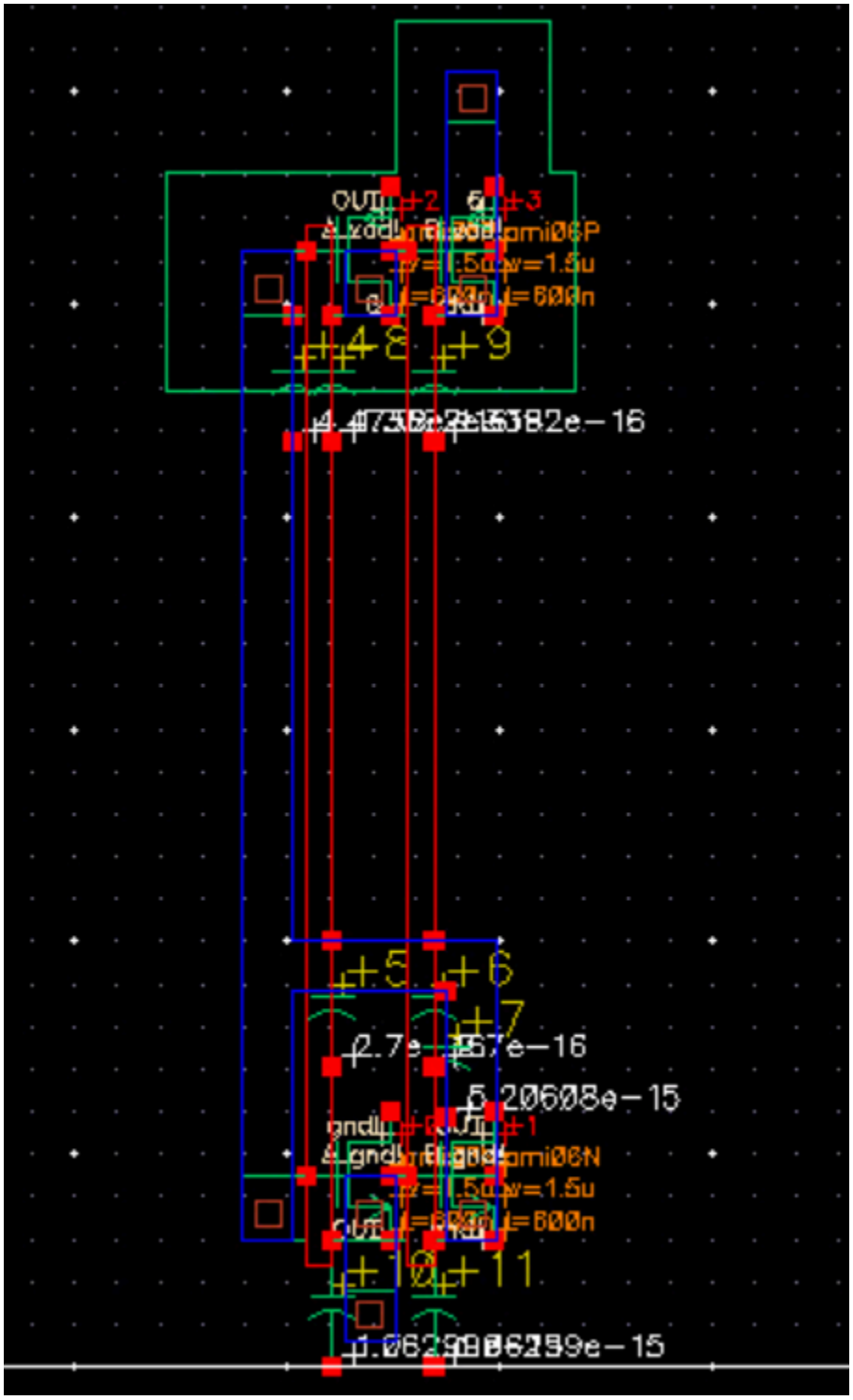
XOR Gate



OR Gate


10
3.66678e-15
pmi06P
w=1.5u
l=600n
13
15
2e-16
5
pmi06P
w=1.5u
l=600n
OUT
8
9
e-16
12
1.63056e-15
6
7
4.917e-16
7.9656e-15
OUT
2
pmi06N
w=1.5u
l=600n
pmi06N
w=1.5u
l=600n
18
14
16
5.0264e-16
e-15

NOR Gate



6. Specification Sheet

- Key specifications of the final design.

  Final Design without PAD

    Height: 804.6 um

    Width: 855.15 um

  Final Design with PAD

    Height: 1500 um

    Width: 1500 um

  Final Design Number of Gates

    XOR: 105

    NAND: 162

    Inverter: 421

    NOR: 56

  Final Design Number of Transistors

    2,288

  Final Design Size of Each Transistor

    Height: 1.5 um

    Width: 1.5 um

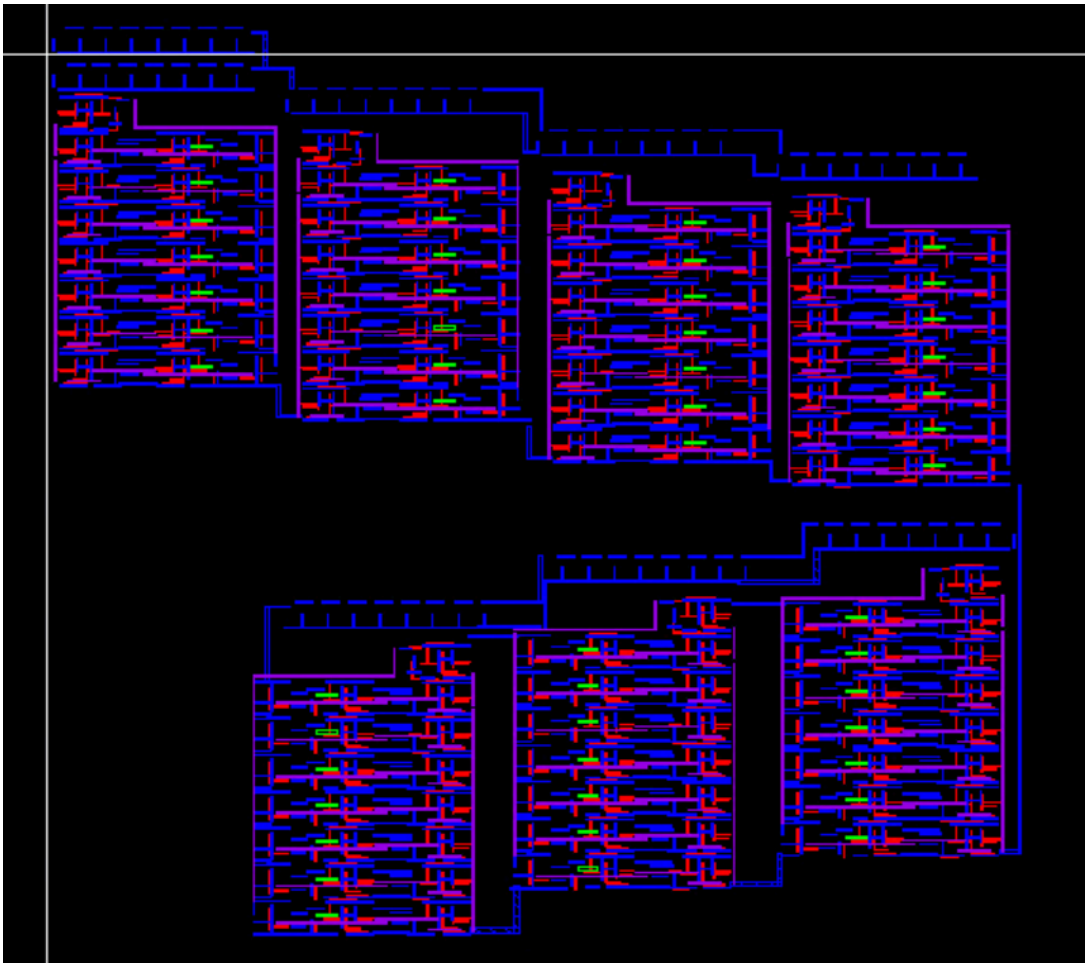
- Pin mapping and functionality table.

  This was not able to be presented due to the expiration of the license within Cadence.

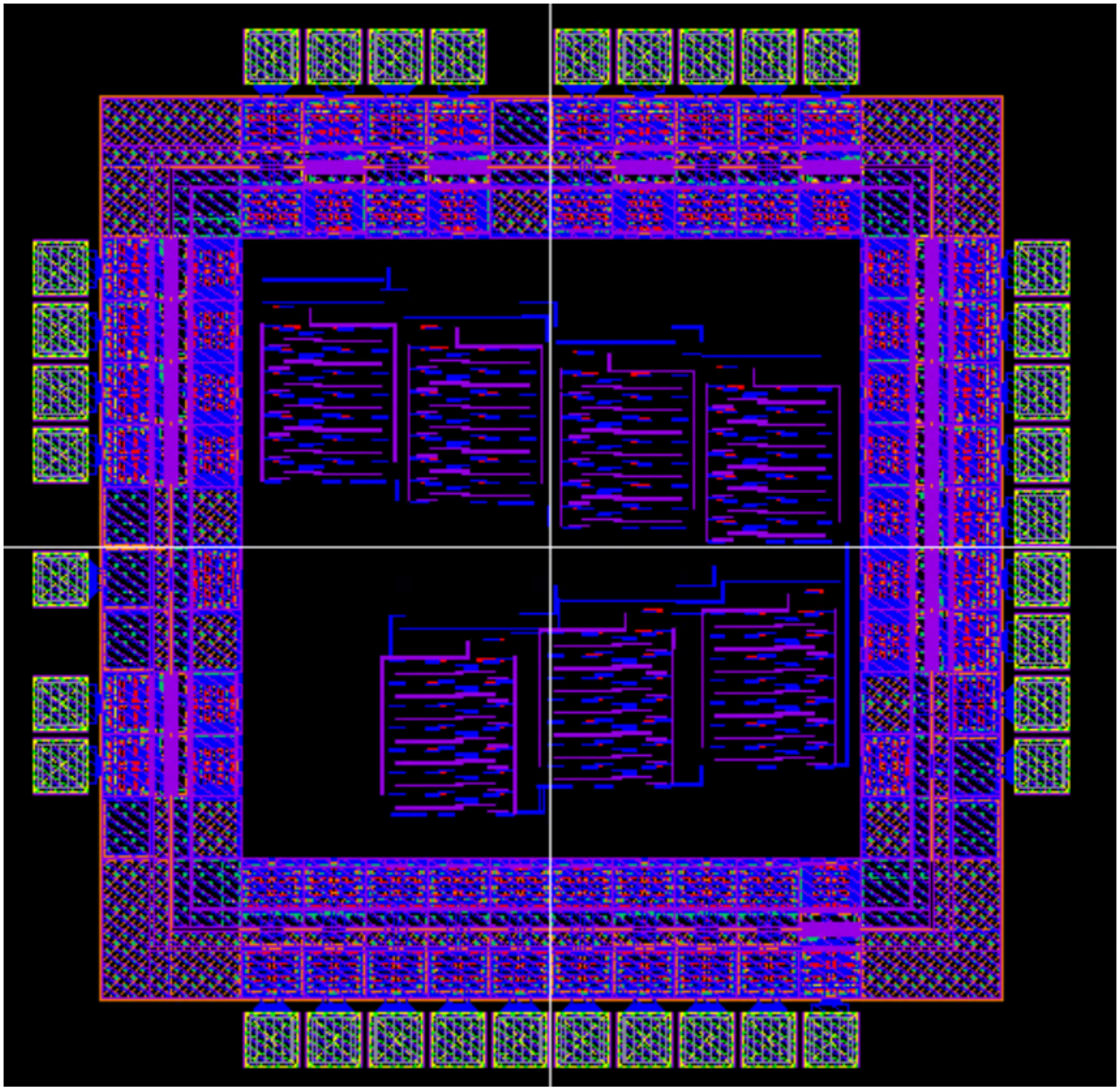
7. Final Layout

- Visual representation of the layout, with and without the pad frame.

Final Layout Without the Pad Frame



Final Layout With the Pad Frame



- Discussion of layout challenges and solutions.

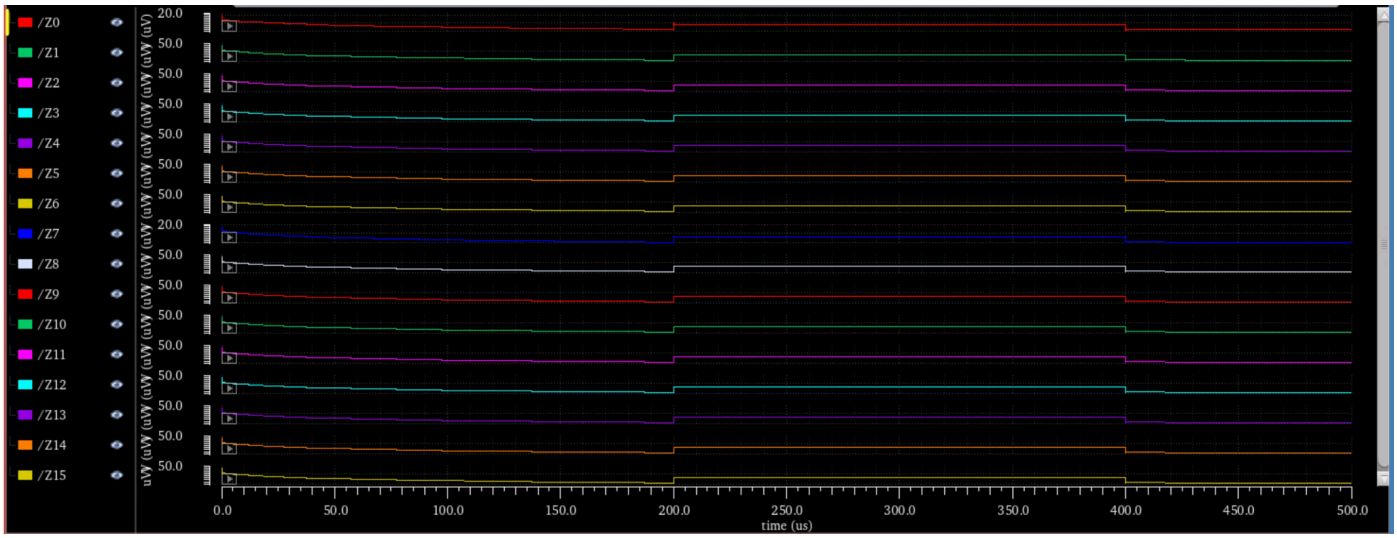
  By far the biggest challenge with creating the layout of the 8-Bit Multiplier was not being able to confirm that the layout matched the schematic, due to the expiration of the license in cadence. It is because of this inability to verify the layout that the pad was not fully connected, because it would not have been efficient or practical to connect the layout in the pad without fully knowing that the layout was valid.

8. Simulations and Verifications

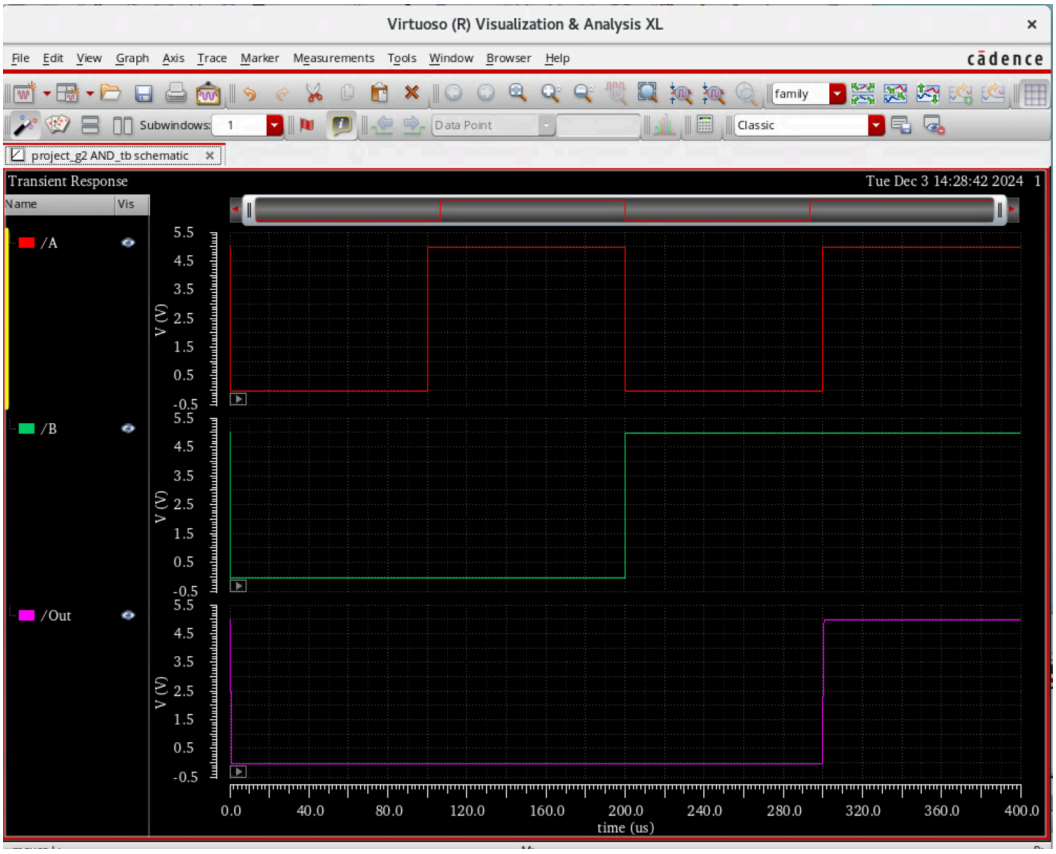
- Overview of simulation results.

  Test bench simulations were performed for the 8-Bit Multiplier and all of the components that made of the project

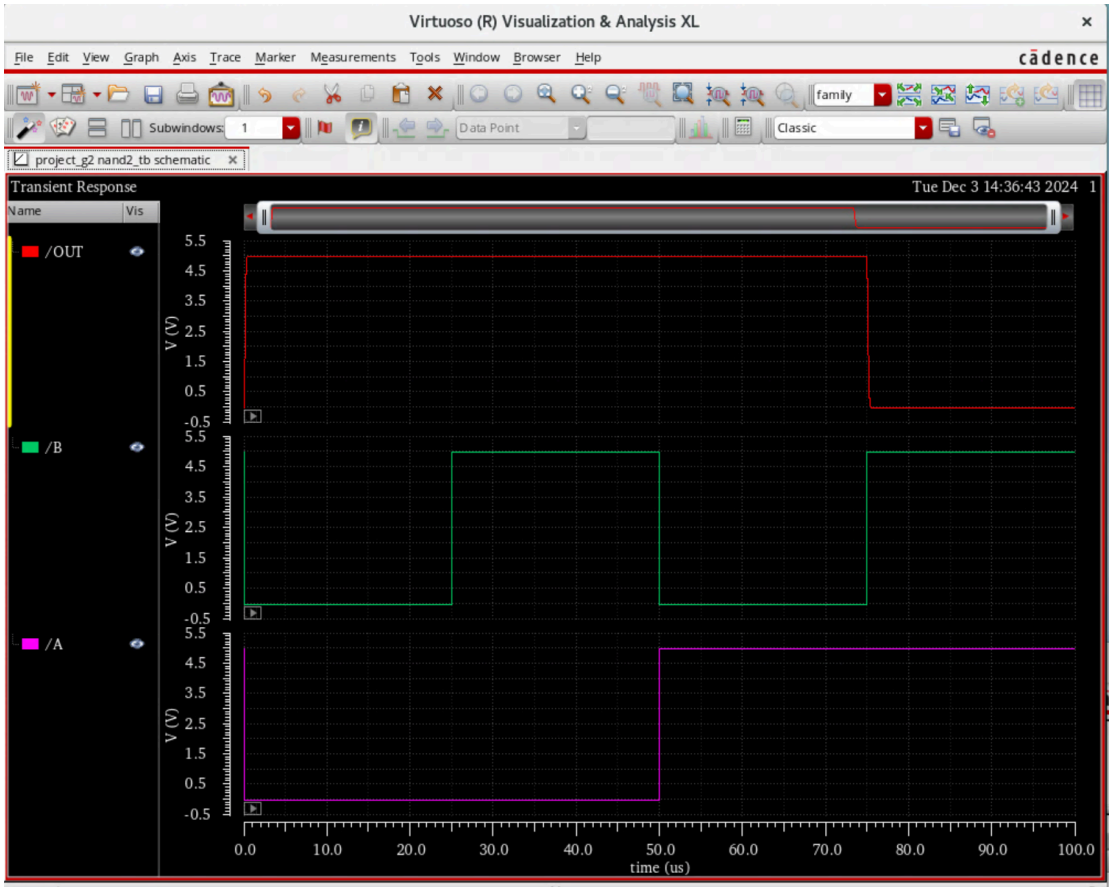
8-Bit Multiplier



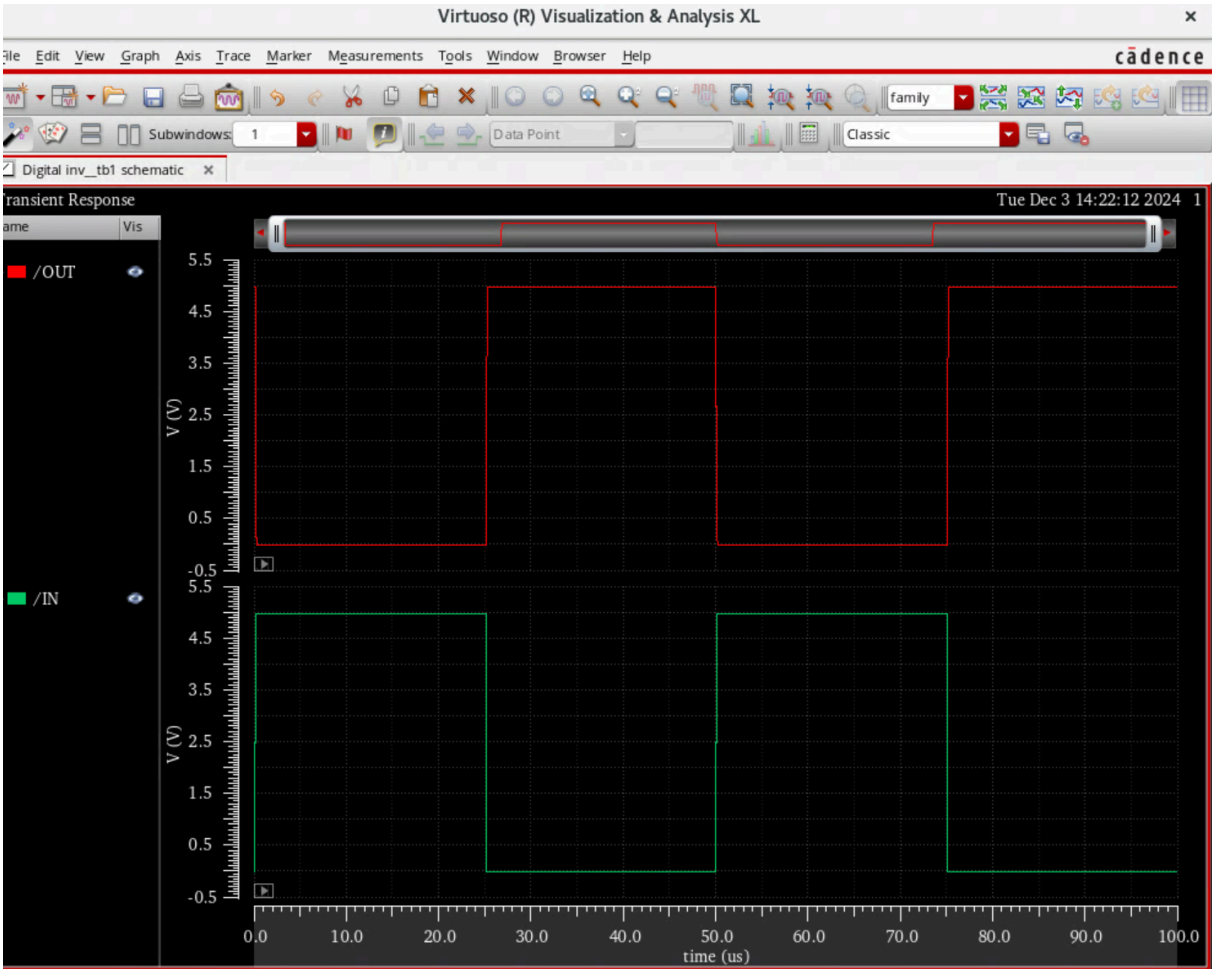
AND Gate


Virtuoso (R) Visualization & Analysis XL
File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help
cadence
Subwindows: 1
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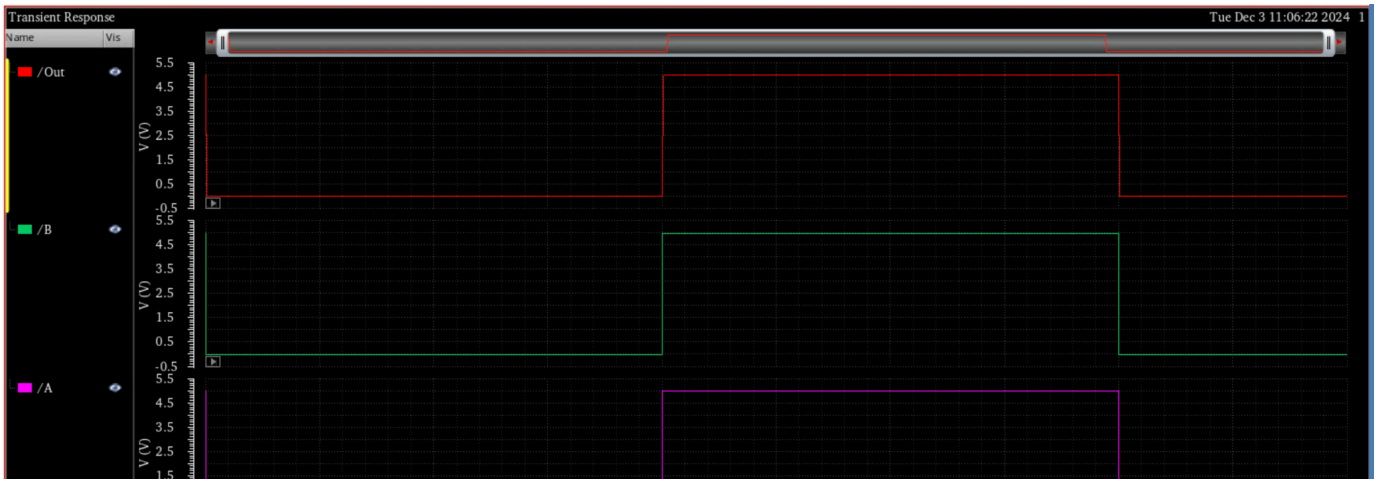
NAND Gate


Virtuoso (R) Visualization & Analysis XL
File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help
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project_g2 nand2_tb schematic
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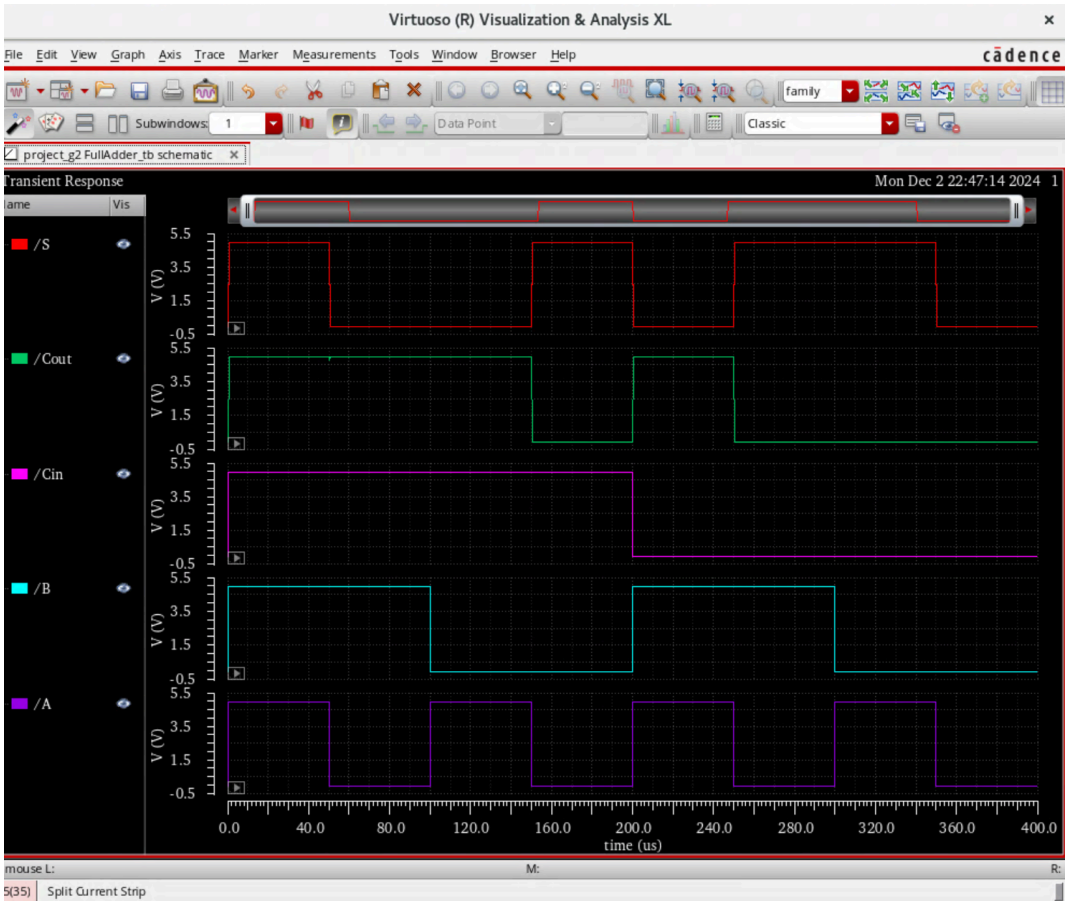
Inverter Gate



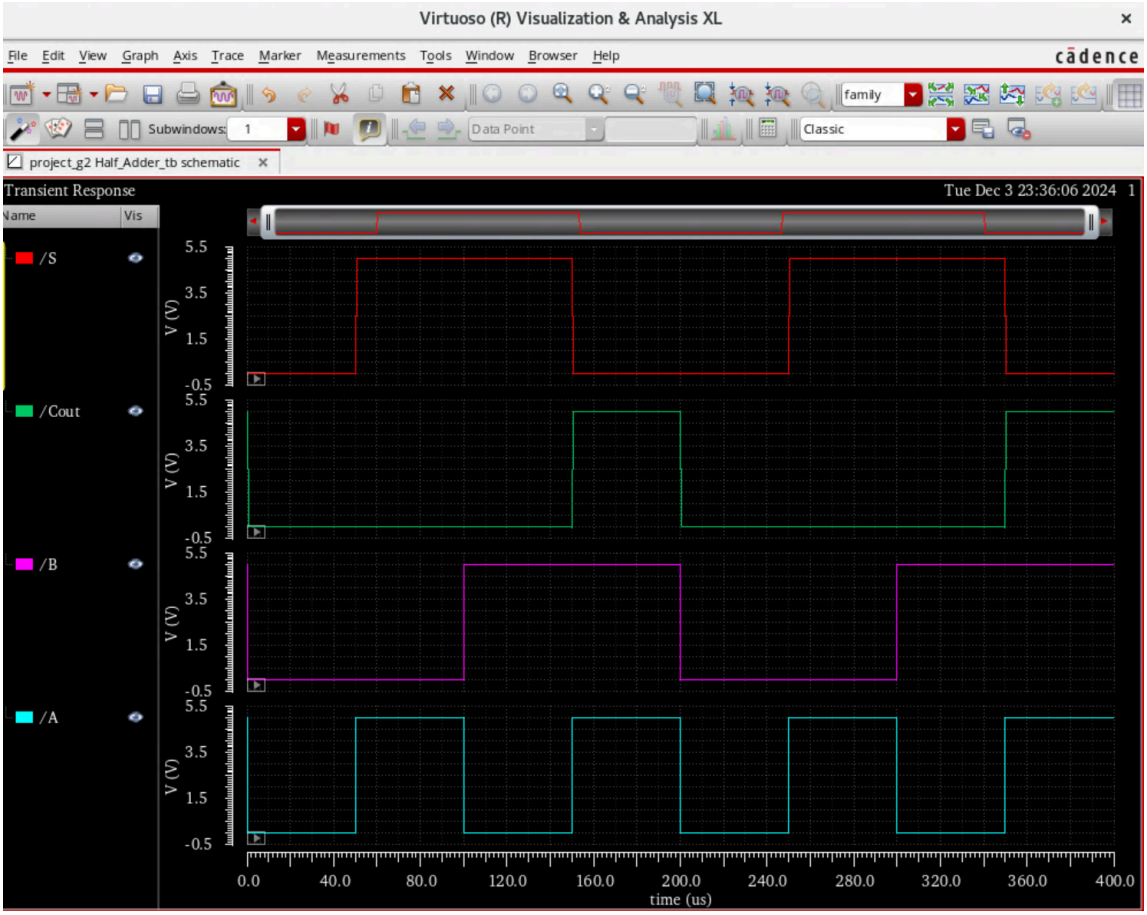
8-Bit Adder



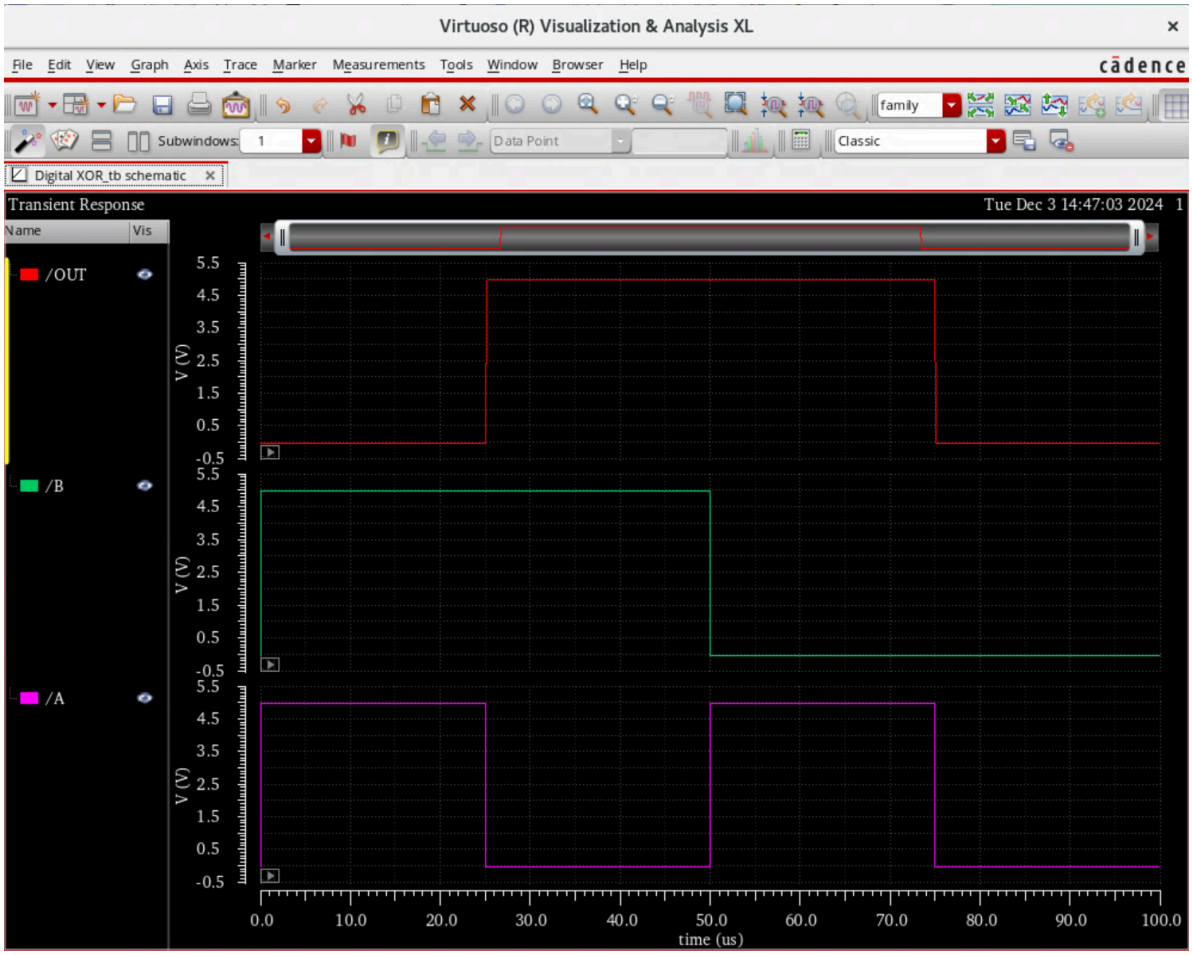
Full Adder



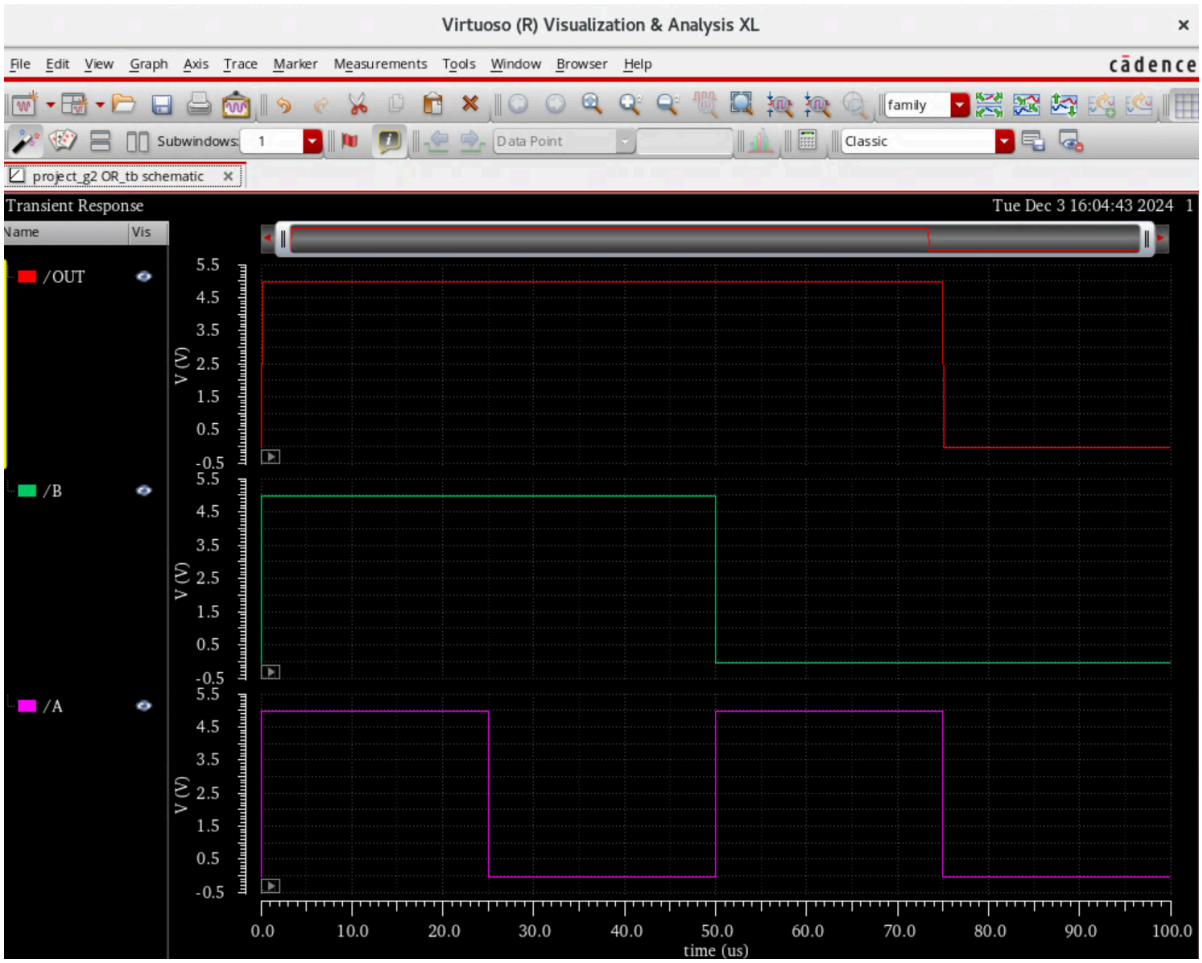
Half Adder


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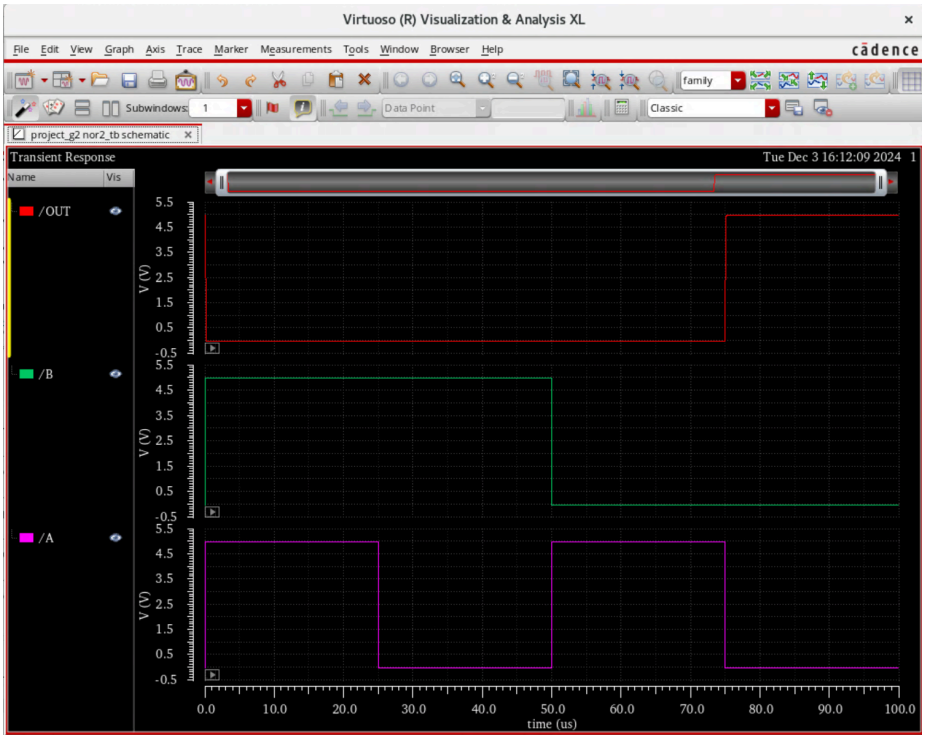
XOR Gate


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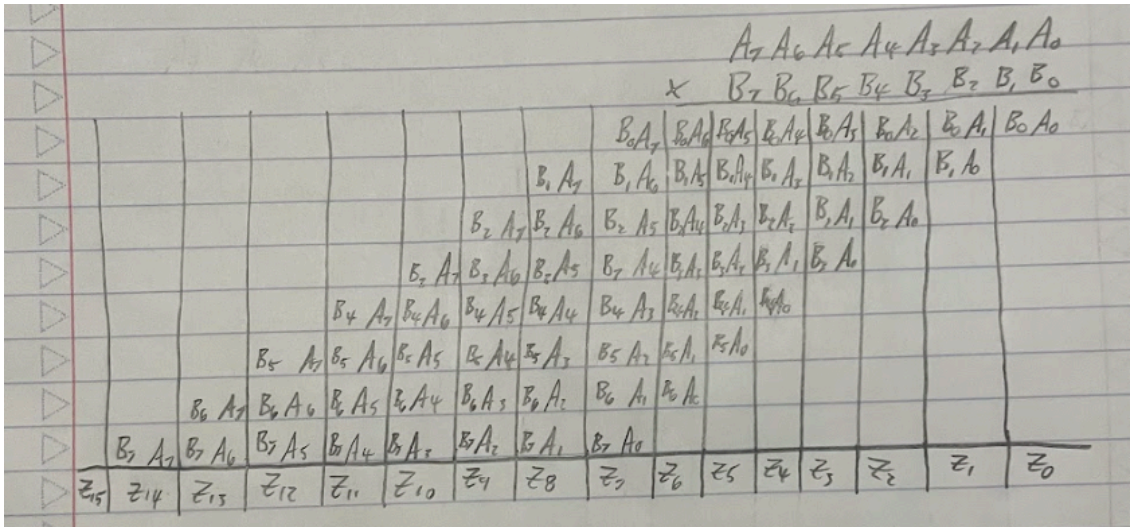
OR Gate


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NOR Gate



- Evidence of the design meeting specifications.



All of the images in this part are evidence of the design meeting the intended specifications. In the image directly above, the functionality of the 8-Bit Multiplier is shown, additionally, in the output graph shown at the start of this section, it can be

seen that the output is representative of the functionality image. Two 8-Bit numbers get multiplied and produce one 16-Bit number. Furthermore, all of the subcomponents that made up this device are validated through the output graphs.

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9. Failure Cases

Throughout the development of the 8-bit multiplier, several failure cases were identified that impacted the design and required iterative solutions. One of the most prominent failure cases occurred in the initial implementation of the 8-bit adder, which utilized two cascaded 4-bit adders. This design exhibited significant timing-related glitches during simulations. Specifically, outputs that were not intended to be active, such as S1,S3,S1,S3, or S4S4 when only S2S2 was expected to be on, displayed small, unintended spikes equal to 11 at the rising and falling edges of the clock. This behavior compromised the integrity of the design, as it introduced transient errors that could propagate through the multiplier and affect the final results.

The root cause of this failure case was traced to improper synchronization between the two cascaded 4-bit adders. The delays inherent in the cascading structure likely caused momentary instability in the outputs, as signals from one stage were not fully settled before being used as inputs for the next. This resulted in race conditions and signal contention, particularly at clock transitions, leading to the observed glitches. Additionally, the carry-out signal (CoutCout) in the initial design experienced significant distortion, as it was generated either too early or too late relative to the clock cycle. This issue further hindered the accuracy of the adder, as CoutCout plays a critical role in propagating carry bits in multi-bit addition.

Another failure case involved the layout and verification of the final design. Although all subcomponents, including AND gates and full adders, passed Design Rule Checks (DRC) and Layout Versus Schematic (LVS) verification individually, the overall layout verification for the 8-bit multiplier was not completed. This was due to an unexpected expiration of the design tool license, which occurred before the final extracted view could be generated. While this failure was external and unrelated to the design itself, it highlighted the importance of anticipating logistical challenges during project planning.

The impacts of these failure cases were significant but ultimately instructive. The timing-related issues in the initial 8-bit adder design delayed progress and required the

development of a more robust solution. By redesigning the adder to use seven full adders and one half adder, these failures were resolved, resulting in a stable and reliable circuit. The license expiration prevented complete verification of the final layout, leaving room for uncertainty about the full functionality of the 8-bit multiplier. However, the successful verification of all subcomponents provides confidence that the final design would have met functional and performance requirements if fully verified. These failure cases emphasized the importance of thorough testing, iterative design improvements, and proactive planning to address potential logistical constraints.

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10. Debugging and Problem Resolution

The debugging and problem resolution phase of the project was pivotal in addressing challenges encountered during the design and implementation of the 8-bit multiplier. A significant issue arose with the initial design of the 8-bit adder, which was implemented using two 4-bit adders cascaded together. While this approach was conceptually sound, it failed in practice due to timing inconsistencies and signal distortion. The carry propagation between the two 4-bit adders introduced significant delays, leading to incorrect output signals and rendering the design unsuitable for integration into the multiplier.

To resolve this issue, the adder design was completely restructured. The second iteration employed a combination of seven full adders and one half adder, allowing for more efficient and accurate carry propagation. This design provided better control over the timing and minimized signal distortion observed in the initial implementation. Each full adder was responsible for one bit of the addition process, while the half adder handled the least significant bit, which required no carry-in. The revised design not only corrected the timing issues but also demonstrated consistent and accurate functionality during simulations.

The debugging process involved a systematic approach to identifying and addressing the root cause of the adder's failure. Initial troubleshooting focused on isolating the carry propagation delays, which were identified as the primary source of the problem. Simulations were run to compare the theoretical and practical behaviors of the 4-bit adder design, revealing discrepancies in the output signals. After redesigning the adder, extensive testing was conducted to ensure the new implementation met the required functional and timing specifications. This iterative process of debugging and refinement was instrumental in overcoming the challenges and ensuring the integrity of the 8-bit multiplier.

Overall, the debugging phase underscored the importance of adaptability and thorough testing in digital design. While the need to redesign the 8-bit adder delayed progress, it also provided valuable insights into carry propagation, timing optimization, and the practical limitations of cascading components. These lessons not only contributed to the success of the multiplier but also highlighted the iterative nature of problem-solving in complex hardware projects.

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11. Timing Analysis

The timing analysis of the project revealed critical insights into the performance and stability of the 8-bit multiplier, particularly during the simulation of the initial 8-bit adder design. One major issue observed was a logic failure in the output signals that were not intended to be active. For instance, if only output S2S2 was supposed to be on, the other outputs would exhibit small, unexpected spikes equal to 11 at the rising and falling edges of the clock. This glitch, occurring in the initial schematic using two cascaded 4-bit adders, pointed to timing inconsistencies in the design. These brief, unintended activations suggested improper synchronization between the cascaded components, causing transient errors as signals propagated through the circuit.

Another significant timing issue was encountered with the carry-out signal (CoutCout), which experienced severe distortion. This distortion appeared to stem from improper signal timing, with CoutCout being generated either too early or too late relative to the expected clock cycle. As CoutCout is critical for propagating carry bits in multi-bit addition, this distortion introduced substantial errors in the final outputs of the multiplier. The inconsistencies in timing not only reduced the accuracy of the circuit but also revealed limitations in the cascading approach of the initial design.

A potential explanation for these issues lies in the nature of the cascaded 4-bit adder implementation. In such designs, the output of one adder directly drives the inputs of the next, creating a dependency chain that amplifies minor delays or mismatches in signal timing. Small timing mismatches can accumulate as signals propagate through the chain, resulting in glitches or distorted outputs at the next clock edge. Additionally, the transitions at the clock edges may have introduced transient glitches due to race conditions, where the signals were momentarily unstable as the circuit adjusted to new input values. These effects are particularly pronounced in cascaded designs, as the cumulative delay can exacerbate synchronization challenges.

In the second iteration of the 8-bit adder, these timing issues were successfully resolved. By redesigning the adder to use seven full adders and one half adder, the propagation of signals was streamlined, eliminating the glitches in inactive outputs. This revised design ensured that only the intended outputs were active during simulations, with no spurious spikes. Additionally, the timing of the CoutCout signal was corrected, aligning it accurately with the clock edges. The improved synchronization between components not only enhanced the accuracy of the adder but also reduced the largest delay encountered during the simulation.

These timing bottlenecks in the initial design highlight the importance of evaluating potential synchronization issues when cascading components. The issues likely arose from a combination of improper timing propagation, cumulative delay, and race conditions, which were mitigated in the second design through a more balanced and modular approach. This iterative design process not only enhanced the performance of the 8-bit multiplier but also reinforced the importance of thorough timing analysis and signal integrity in digital circuit design.

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12. Design Improvement Analysis

The design and implementation of the 8-bit multiplier involved a detailed analysis of its components, testing, simulations, layout, and verification processes. Central to this design were the full adders and AND gates, which formed the backbone of the multiplier. Each full adder, composed of 34 transistors, was essential for summing partial products and carries. With 56 full adders in the design, the total transistor count for this component reached 1,904. Similarly, the 64 AND gates, each using 6 transistors, contributed an additional 384 transistors. This breakdown illustrates the complexity of the multiplier, requiring a total of 2,288 transistors for its operation, highlighting the trade-offs in area and design complexity for such a circuit.

Testing and simulations confirmed the functionality of the components, with minimal to no timing delays observed during the process. These results validated the design's theoretical correctness and its ability to meet performance requirements under simulated conditions. However, initial issues with the 8-bit adder component revealed some challenges in practical implementation. Specifically, output signals that were not active exhibited significant distortion, deviating from expected simulation results. The carry-out signal (`Cout`) initially failed to register correctly, complicating the verification of the overall multiplier. These issues underscored the importance of robust testing at every stage of the design.

The layout and verification process highlighted the project's successes and limitations. Subcomponent layouts, such as those for AND gates and full adders, were accurately designed and successfully verified through DRC and LVS checks. These verifications ensured the physical designs aligned with their corresponding schematics, providing confidence in the integrity of the individual components. Unfortunately, the final layout verification for the complete 8-bit multiplier could not be completed due to an expired license, which also prevented the extraction of parasitic views. This setback highlighted the dependency on external tools and the importance of contingency planning for tool availability in future projects.

Despite these challenges, the project demonstrated strong adherence to design principles and methodologies. The successful verification of subcomponents suggests that, given sufficient time and tool access, the final layout would have met all verification standards. The analysis of transistor counts and functional behavior also provided valuable insights into resource utilization and performance trade-offs. Overall, the project was a practical learning experience in digital design, emphasizing the need for thorough testing, verification, and resource management to ensure the success of complex hardware designs.

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13. Conclusion

The project presented several significant challenges, particularly during the simulation phase, where resolving timing errors proved to be the most difficult aspect. These errors initially caused delays in achieving a fully functional design and required careful analysis to address. Additionally, ensuring the functionality of all components during the layout process was a demanding task. While the individual subcomponents, such as AND gates and full adders, were successfully verified using Design Rule Check (DRC) and Layout Versus Schematic (LVS) checks, integrating these elements into the complete design introduced complexities.

Despite these efforts, the final layout and extracted view verification for the 8-bit multiplier design could not be completed due to time and licensing constraints. However, the successful verification of all subcomponents instilled confidence that the overall design would have met the required verification standards. This project provided invaluable experience in digital design and verification, emphasizing the importance of meticulous testing and iteration in overcoming technical challenges.